

SWIFT SHIP TRACKER

College Name:Anjalai Ammal Mahalingam

Engineering College

College Code :8204

Dept Name :B.Tech Information Technology

Team ID : SWTID-2026-6127

Team Size : 3

Team Leader : Dhaarani V -

dhaaranivenugopal16@gmail.com

Team member : M Bharathi Sahana –

bharathisa50@gmail.com

Team member : Kavya V –

Kavyavenkatesan305@gmail.com

SWIFT SHIP TRACKER

User Story:

The SwiftShip Tracker is designed to provide a centralized Salesforce-based shipment operations solution that enables logistics teams to monitor shipments, coordinate delivery activities, manage delivery exceptions, and maintain shipment history throughout the delivery lifecycle. The system manages Shipment, Route, Delivery Attempt, Delivery Exception, and Proof of Delivery information using Salesforce custom objects and relationships. Salesforce Flows automate shipment status updates, exception assignments, follow-up activities, and operational notifications, while AgentForce AI and Prompt Builder assist authorized users in retrieving shipment information, summarizing delivery exceptions, and obtaining operational insights through conversational interaction. Delivery agents can record delivery attempts and update shipment milestones, while operations teams can monitor unresolved exceptions, route performance, and delivery activities through Reports and Dashboards. Salesforce security controls provide role-based access to shipment information and ensure that users can access the operational data relevant to their responsibilities.

1.INTRODUCTION

1.1 Project Overview :

The SwiftShip Tracker is a Salesforce-based shipment operations management solution developed to improve the monitoring and coordination of shipment activities throughout the delivery lifecycle. Instead of focusing only on parcel booking and basic tracking, the system concentrates on shipment milestones, route coordination, delivery attempts, delivery exceptions, and proof of delivery. The system maintains centralized information about Shipment, Route, Delivery Attempt, Delivery Exception, and Proof of Delivery using Salesforce custom objects and relationships. Operations teams can monitor shipment progress, while delivery agents can record delivery attempts and report issues encountered during delivery. Salesforce automation helps manage shipment

updates, exception assignments, follow-up activities, and operational notifications. The solution also uses AgentForce AI and Prompt Builder to assist authorized users in retrieving shipment information and understanding delivery exceptions through conversational interaction. Reports and dashboards provide operational visibility into delayed shipments, unresolved exceptions, route activities, and delivery performance.

1.2 Purpose :

The purpose of SwiftShip Tracker is to:

- Centralize shipment, route, delivery attempt, and exception information.
- Monitor shipment milestones throughout the delivery lifecycle.
- Identify and manage delivery exceptions such as delays, failed delivery attempts, and route-related issues.
- Automate exception assignment, follow-up activities, and important operational notifications.
- Maintain a structured history of delivery attempts and proof-of-delivery information.
- Enable conversational access to shipment and exception information using AgentForce AI and Prompt Builder.
- Improve operational efficiency through Salesforce Flows and automated processes.
- Provide reports and dashboards for monitoring shipment status, route activity, unresolved exceptions, and delivery performance.
- Provide secure, role-based access to operations managers, dispatch coordinators, delivery agents, support staff, and administrators.

2. IDEATION PHASE

2.1 Problem Statement (Operations Problem Statement Template)

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1 Operations Manager	Operations Manager	identify delayed and problematic shipments	shipment issues may be recorded across different operational activities	shipment information and exception details are not centrally monitored	concerned
PS-2 Dispatch Coordinator	Dispatch Coordinator	coordinate shipments with delivery routes	route and shipment information may require separate monitoring	delivery activities need better coordination and visibility	pressured
PS-3 Delivery Agent	Delivery Agent	report unsuccessful delivery attempts and delivery problems	exception reporting may require manual communication	delivery issues need to be recorded and assigned systematically	concerned
PS-4 Operations Staff	Operations Staff	resolve shipment exceptions quickly	unresolved exceptions may remain without proper follow-up	there is a need for automated assignment and escalation	frustrated
PS-5 Operations Manager	Operations Manager	monitor route and delivery performance	operational information is difficult to analyze manually	management needs centralized reports and dashboards	uncertain
PS-6 Delivery Agent	Delivery Agent	maintain accurate delivery attempt records	previous delivery activities may not be organized in one place	shipment history needs to be maintained systematically	concerned
PS-7 Support Staff	Support Staff	obtain shipment and exception information quickly	searching through multiple records can take time	operational information should be easily accessible	frustrated
PS-8 Administrator	Administrator	control access to shipment information	different users have different responsibilities	shipment data requires appropriate role-based security	responsible

PS	I am (User)	I'm trying to...	But...	Because...	Which makes me feel...
PS-1	 Operations Manager	 identify delayed and problematic shipments	 shipment issues may be recorded across different operational activities	 shipment information and exception details are not centrally monitored	 concerned
PS-2	 Dispatch Coordinator	 coordinate shipments with delivery routes	 route and shipment information may require separate monitoring	 delivery activities need better coordination and visibility	 pressured
PS-3	 Delivery Agent	 report unsuccessful delivery attempts and delivery problems	 exception reporting may require manual communication	 delivery issues need to be recorded and assigned systematically	 concerned
PS-4	 Operations Staff	 resolve shipment exceptions quickly	 unresolved exceptions may remain without proper follow-up	 there is a need for automated assignment and escalation	 frustrated
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PS-8	 Administrator	 control access to shipment information	 different users have different responsibilities	 shipment data requires appropriate role-based security	 responsible



Solving these challenges with Salesforce to create smarter shipment operations, exceptional delivery experiences, and stronger business outcomes.



*Moving Possibilities
Together*

2.2 Empathy Map Canvas

User Persona: Operations Manager

- **Think & Feel:** Wants a clear and centralized view of shipment activities, delayed shipments, delivery exceptions, and unresolved operational issues so that problems can be addressed quickly.
- **Hear:** Receives updates from dispatch coordinators, delivery agents, support staff, and other operations personnel regarding shipment delays, delivery attempts, and exceptions.
- **See:** Shipment status, route information, delivery attempts, exception records, resolution status, proof-of-delivery information, reports, and operational dashboards.
- **Say & Do:** Monitors shipment operations, reviews delayed shipments, assigns or follows up on delivery exceptions, checks route performance, and reviews operational reports.

- **Pain:** Difficulty identifying delayed shipments, scattered exception information, manual follow-up, unresolved delivery issues, and limited visibility into overall shipment performance.

- **Gain:** Centralized shipment visibility, automated exception handling, timely alerts, clear operational dashboards, organized delivery history, and faster identification of shipments requiring attention.

Swift Ship Tracker Empathy Map

User Persona: Operations Manager



Think & Feel

- Wants a clear and centralized view of all shipment activities, delayed shipments, delivery exceptions, and unresolved issues.
- Feels responsible for smooth operations and on-time deliveries.
- Wants to solve problems quickly and keep customers satisfied.



Hear

- Receives regular updates from dispatch coordinators, delivery agents, customer support staff, and other operations personnel.
- Hears about shipment delays, delivery attempts, route issues, and exceptions.
- Gets feedback from customers regarding delivery status and service quality.



See

- Shipment statuses, route information, and delivery attempts.
- Exception records, resolution status, and proof-of-delivery information.
- Operational reports, analytics, and performance dashboards.



Say & Do

- Monitors daily shipment operations.
- Reviews delayed shipments and tracks delivery exceptions.
- Coordinates with teams for follow-ups and resolves issues.
- Checks route performance and verifies delivery records.
- Looks at operational reports to improve efficiency.



Operations Manager



Pain

- Difficulty identifying delayed shipments.
- Scattered exception information across multiple channels.
- Manual follow-ups and time-consuming coordination.
- Unresolved delivery issues and repeat exceptions.
- Limited visibility into overall shipment performance.



Gain

- Centralized shipment visibility.
- Automated exception tracking and handling.
- Timely alerts for delayed shipments and issues.
- Clear operational dashboards and reports.
- Organized delivery history and proof-of-delivery records.
- Faster identification of shipments requiring attention.

2.3 Brainstorming & Idea Prioritization

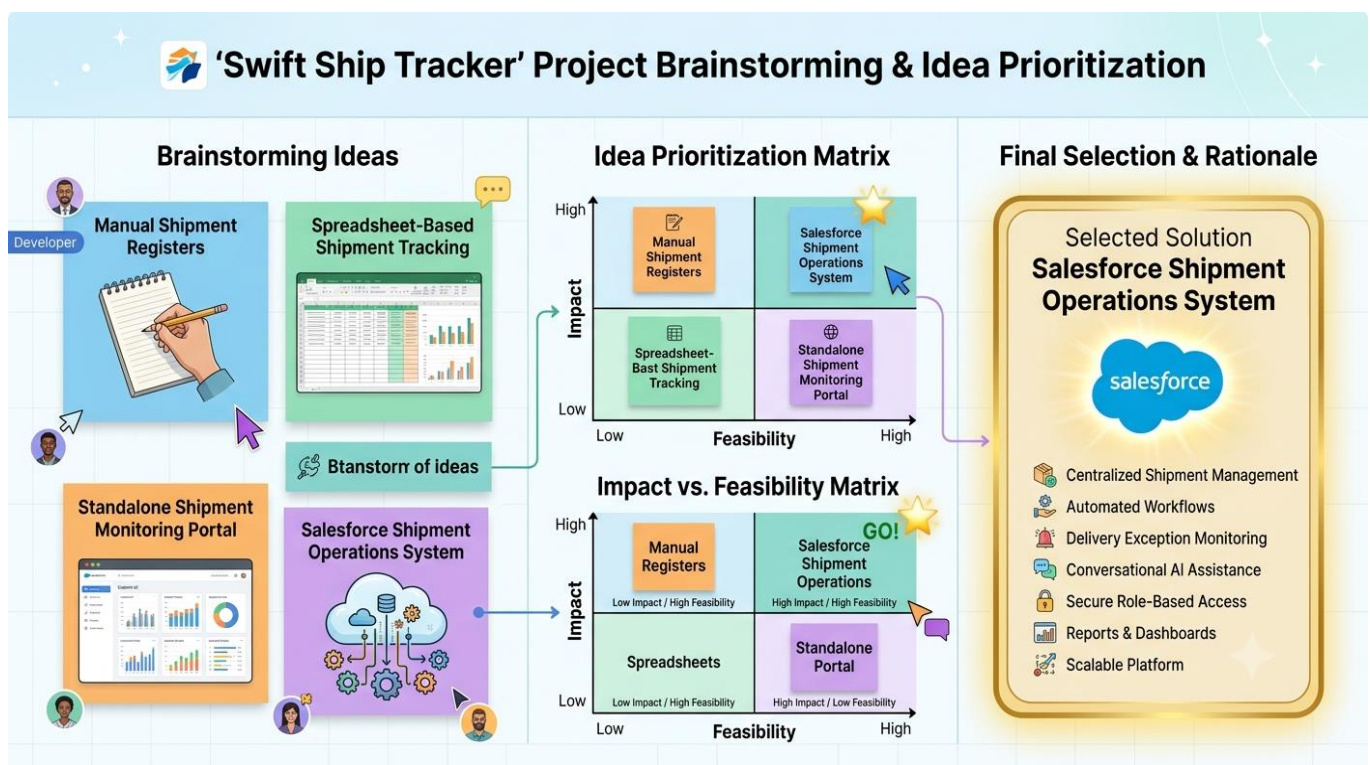
Ideas Considered

- Manual Shipment Registers
- Spreadsheet-Based Shipment Tracking
- Standalone Shipment Monitoring Portal
- Salesforce Shipment Operations System

Final Selection

Salesforce-based Shipment Operations System was selected because it provides:

- Centralized shipment and exception management
- Automated operational workflows
- Delivery exception monitoring
- Conversational AI assistance
- Secure role-based access
- Reports and dashboards
- Scalable Salesforce platform



3. REQUIREMENT ANALYSIS PHASE

3.1 Shipment Operations Journey Map

Stages:

- Shipment Creation
- Route Assignment
- Shipment Dispatch
- In-Transit Monitoring
- Delivery Attempt
- Exception Identification
- Exception Assignment & Resolution
- Proof of Delivery
- Shipment Completion
- Operational Review & Reporting

Shipment Operations Flow:

Shipment Creation



Route Assignment



Dispatch



In-Transit Monitoring



Delivery Attempt



Successful Delivery / Exception



Exception Assignment & Resolution



Proof of Delivery



Shipment Completion

SwiftShip Tracker Shipment Operations Journey Map



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Shipment record creation and management
FR-2	Route assignment and route information management
FR-3	Shipment milestone and status tracking
FR-4	Delivery attempt recording and history management
FR-5	Delivery exception creation and classification
FR-6	Exception severity and resolution status management
FR-7	Automatic assignment and escalation of critical exceptions
FR-8	Proof-of-delivery information management
FR-9	Automated operational notifications and follow-up tasks
FR-10	AI-assisted shipment and exception information retrieval
FR-11	Reports and dashboards for shipment and operational performance
FR-12	Secure role-based access for operations users and delivery agents

Non-Functional Requirements

NFR	Description
Usability	Provide an intuitive Salesforce interface for operations, dispatch, support, and delivery users.

Security	Provide role-based, object-level, and field-level access to shipment and operational information.
Reliability	Maintain accurate shipment status, delivery attempt, exception, and resolution information.
Performance	Provide timely Flow automation, record updates, notifications, and AgentForce responses.
Availability	Provide cloud-based access to authorized users through the Salesforce platform.
Scalability	Support increasing numbers of shipments, routes, delivery attempts, and exceptions.
Data Integrity	Maintain consistent relationships between shipments, routes, delivery attempts, exceptions, and proof-of-delivery records.
Maintainability	Allow administrators and developers to update Flows, Apex logic, prompts, AgentForce topics, permissions, and configurations when required.

3.3 Data Flow Diagram

Main Operational Data Flow

The SwiftShip Tracker manages shipment information throughout the operational delivery lifecycle.

Operations User



Shipment Creation



Shipment Record



Route Assignment



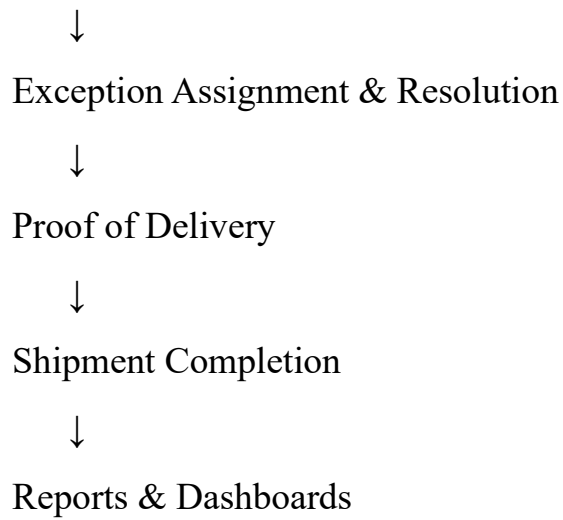
Dispatch & In-Transit Monitoring



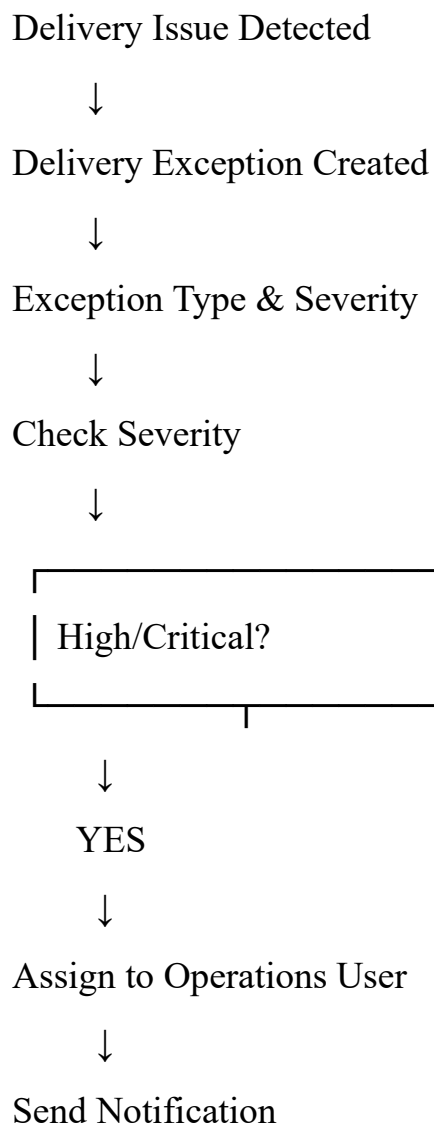
Delivery Attempt

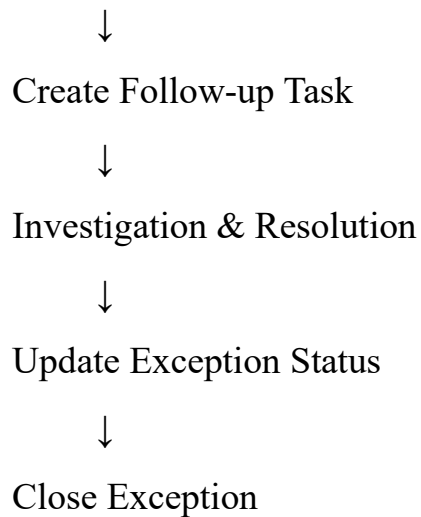


Successful Delivery / Delivery Exception

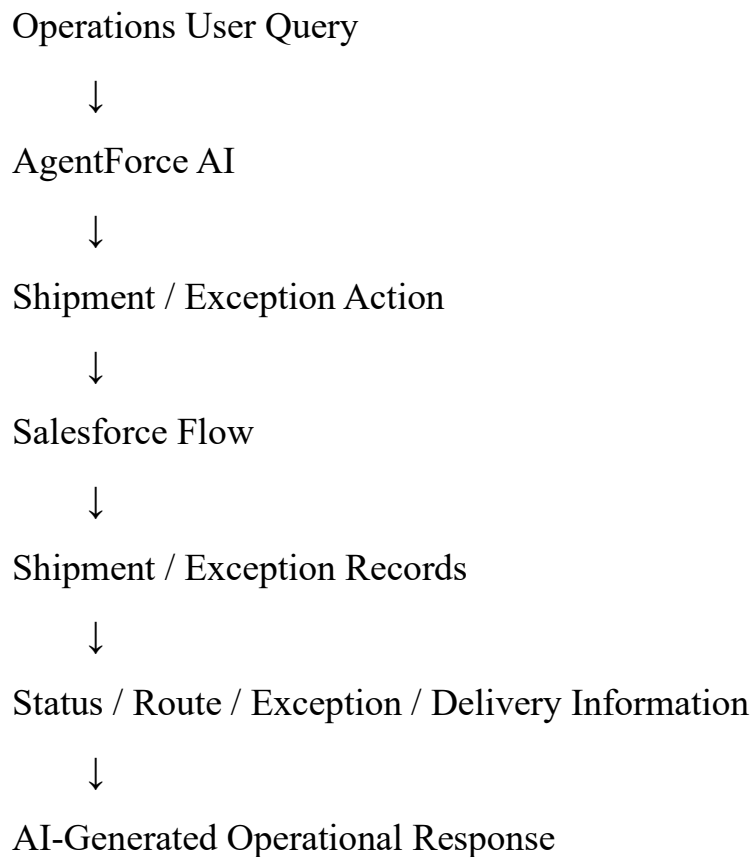


Exception Management Flow

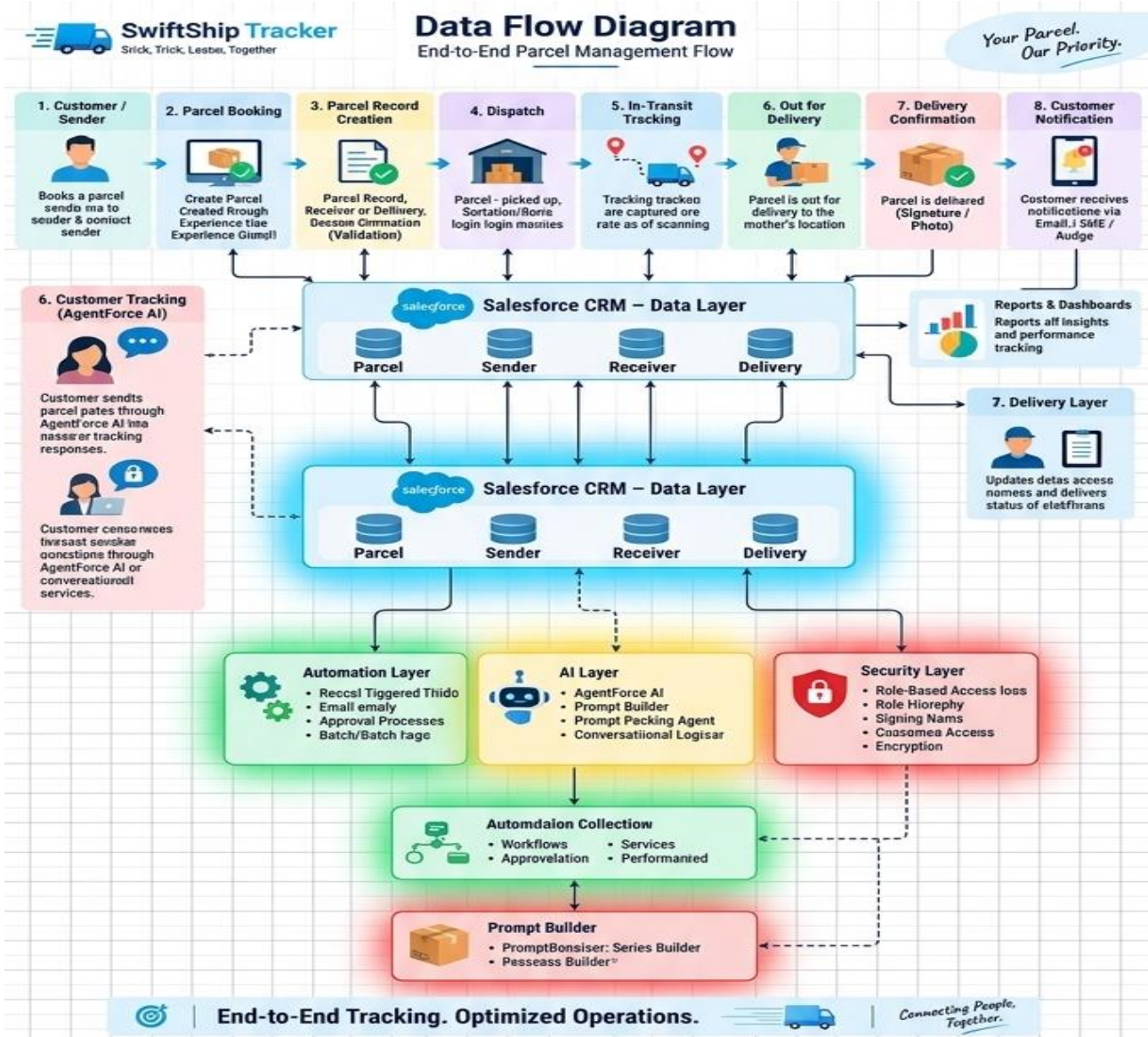




AI Operations Flow



The AI component is designed to assist authorized users with operational questions rather than simply retrieving basic parcel details.



3.4 Technology Stack

Layer	Technology
Platform	Salesforce Developer Edition
UI	Lightning App Page, Lightning Record Pages
Database	Salesforce Custom & Standard Objects
Business Logic	Salesforce Flows, Apex Classes
Automation	Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows
AI	AgentForce AI, Prompt Builder
Notifications	Email Alerts, Email Templates, Salesforce Tasks

Processing	Apex processing where required
Security	Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
Reporting	Reports & Dashboards
Operational Data	Shipment, Route, Delivery Attempt, Delivery Exception, Proof of Delivery objects

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Shipment information is difficult to monitor centrally	Salesforce Shipment Management	Centralized shipment visibility
Route and shipment activities require separate monitoring	Route and Shipment Objects	Connected operational information
Delivery attempts may not be systematically recorded	Delivery Attempt Management	Organized delivery history
Delivery problems may not be recorded consistently	Delivery Exception Management	Structured issue tracking
Critical exceptions require quick attention	Flow-based Assignment & Escalation	Faster operational follow-up
Manual follow-up can cause unresolved issues	Automated Tasks and Notifications	Timely action on exceptions
Delivery completion information needs to be maintained	Proof of Delivery Management	Structured delivery confirmation
Shipment information can take time to retrieve	AgentForce AI + Prompt Builder	Conversational operational assistance
Operational performance is difficult to analyze	Reports & Dashboards	Better operational visibility
Different users have different responsibilities	Profiles, Roles & Permission Sets	Controlled access
Shipment history can become difficult to maintain	Related Salesforce Objects	Connected and traceable records
Management needs to identify problem areas	Exception and Route Reports	Improved operational analysis

Swift Ship Tracker Problem – Solution Fit

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Right Problems. Smarter Solutions. A Better Delivery Experience.



Moving Possibilities Together

4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Difficult and inefficient monitoring of shipment operations, delivery attempts, routes, and delivery exceptions
Solution	Salesforce-based Shipment Operations and Exception Management System
Innovation	AgentForce AI + Prompt Builder for conversational shipment and exception assistance, supported by automated Salesforce Flows
Impact	Centralized shipment monitoring, organized exception handling, automated follow-up, improved delivery visibility, and operational reporting
Business Model	Shipment, route, delivery-agent, operations, support, and administrator operations
Scalability	Supports growing shipment operations, additional routes, delivery activities, exception records, dashboards, and future integration with external logistics systems

4.3 Solution Architecture

- Users

Operations Manager | Dispatch Coordinator | Delivery Agent | Support Staff | Admin



- Salesforce Lightning Platform

Lightning App Page | Lightning Record Pages



- SwiftShip Tracker Salesforce Platform

- Data Layer

- Shipment
- Route
- Delivery Attempt
- Delivery Exception
- Proof of Delivery



- Automation Layer

- Record-Triggered Flows
- Auto-Launched Flows
- Scheduled Flows
- Apex Classes
- Operational Tasks
- Email Alerts



- AI Layer

- AgentForce AI
- Prompt Builder
- Shipment Operations Assistant
- Exception Summary Action



- Security Layer

- Profiles
- Roles
- Permission Sets
- Sharing Rules
- Field-Level Security



- Reports & Dashboards



- Shipment Operations & Exception Insights

Architecture Flow

Users



Salesforce Lightning



SwiftShip Tracker



Data Layer



Automation + Apex



AgentForce + Prompt Builder



Security Controls

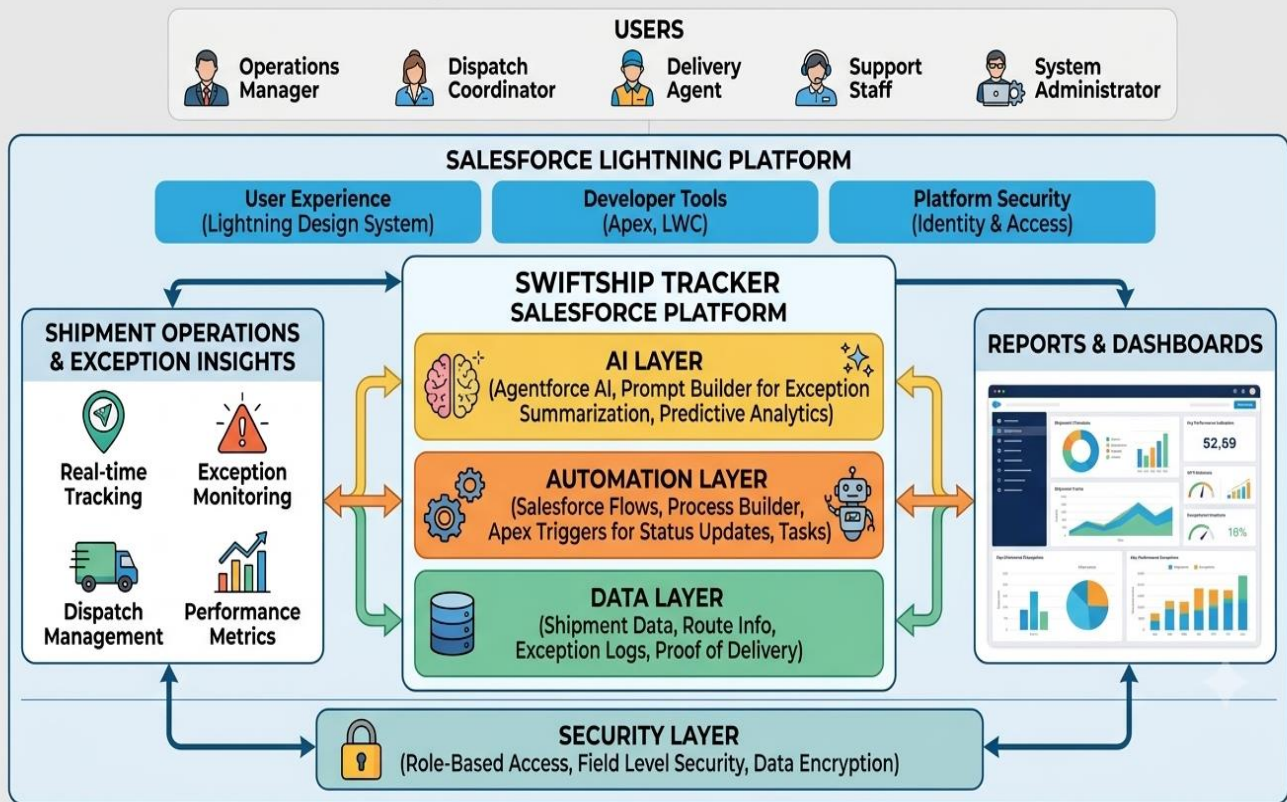


Reports & Dashboards



Shipment Operations & Exception Insights

SwiftShip Tracker Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → User Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

The project is divided into multiple sprints so that the shipment management system can be developed progressively from Salesforce environment setup and data modeling to automation, AI, reporting, testing, and final documentation.

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure the Salesforce environment so that I can build the SwiftShip Tracker shipment operations system.	3	High	Member 1
Sprint-2	Data Modeling	USN-2	As an administrator, I want to create Shipment, Route, Delivery Attempt, Delivery Exception, and Proof of Delivery objects with appropriate relationships so that shipment operations can be stored systematically.	5	High	Member 2
Sprint-2	Data Modeling	USN-3	As an operations user, I want dedicated tabs and a Lightning App for SwiftShip modules so that I can easily access shipment and operational records.	5	High	Member 3
Sprint-3	Shipment Operations	USN-4	As a dispatch coordinator, I want to assign shipments to routes and update shipment milestones so that shipment movement can be monitored throughout the delivery lifecycle.	5	High	Member 4
Sprint-3	Exception Management	USN-5	As a delivery agent, I want to record delivery attempts and create delivery exceptions so that unsuccessful or	5	High	Member 4

			problematic deliveries are properly documented.			
Sprint-4	Automation & Apex	USN-6	As a system, I want automated Flows to assign critical exceptions, create follow-up tasks, and send notifications so that operational issues receive timely attention.	5	High	Member 1
Sprint-4	Automation & Apex	USN-7	As an operations system, I want Apex processing to evaluate shipment performance and exception-related information so that complex operational processing can be handled accurately.	5	High	Member 2
Sprint-5	Reports	USN-8	As an operations manager, I want reports showing shipment status, delivery attempts, route activity, and unresolved exceptions so that I can monitor operational performance.	4	High	Member 3
Sprint-6	Dashboards & AI	USN-9	As an operations manager, I want dashboards showing shipment and exception metrics while AgentForce provides conversational operational assistance so that I can review shipment activities efficiently.	5	Medium	Member 4
Sprint-7	Testing & Documentation	USN-10	As a project team member, I want to test the Salesforce configuration, automation, security, reports, and AI	3	High	Member 1

			functionality so that the final SwiftShip Tracker system is properly validated and documented.			
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5.2 Project Tracker, Velocity & Burndown Chart

Spr int	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	3	6 Days	07 Sep 2026	12 Sep 2026	3	12 Sep 2026
Sprint-2	10	6 Days	14 Sep 2026	19 Sep 2026	10	19 Sep 2026
Sprint-3	10	6 Days	21 Sep 2026	26 Sep 2026	10	26 Sep 2026
Sprint-4	10	6 Days	28 Sep 2026	03 Oct 2026	10	03 Oct 2026
Sprint-5	4	6 Days	05 Oct 2026	10 Oct 2026	—	—
Sprint-6	5	6 Days	12 Oct 2026	17 Oct 2026	—	—
Sprint-7	3	6 Days	19 Oct 2026	24 Oct 2026	—	—

6. PROJECT DEVELOPMENT PHASE

Phase 1: Requirement Analysis & Planning

Objective

Operations teams often face difficulty monitoring shipment progress, coordinating delivery activities, and resolving delivery exceptions because shipment, route, delivery-attempt, and exception information may not be centrally organized. A Salesforce-based operational system can simplify this process by bringing shipment monitoring, exception management, automated follow-up, and operational insights into one platform.

Key Requirements

- Requirement gathering from operations managers, dispatch coordinators, delivery agents, and support staff.
- Defining system scope and user personas such as **Operations Manager, Dispatch Coordinator, Delivery Agent, Support Staff, and Admin.**
- Designing the data model with relationships between **Shipment, Route, Delivery Attempt, Delivery Exception, and Proof of Delivery.**
- Identifying use cases such as shipment creation, route assignment, milestone tracking, delivery-attempt recording, exception management, and proof-of-delivery recording.
- Selecting Salesforce tools such as **Salesforce Flows, Apex, Prompt Builder, AgentForce AI, Reports, and Dashboards.**

Defining Project Scope & Objectives

Project Scope

1. Build a scalable Salesforce solution for logistics operations to manage **shipments, routes, delivery attempts, exceptions, and proof of delivery.**
2. Support the shipment lifecycle from **shipment creation → route assignment → dispatch → in-transit monitoring → delivery attempt → exception handling → delivery completion.**
3. Include structured data modeling, Salesforce automation, Apex processing where required, AI-assisted operations, and reporting dashboards.
4. Provide role-based access for **operations managers, dispatch coordinators, delivery agents, support staff, and administrators.**

5. Provide centralized operational visibility into shipment progress, delivery exceptions, route activities, and unresolved issues.

Objectives Summary

- Streamline shipment operations by automating status updates, exception assignment, follow-up tasks, and notifications.
- Improve operational visibility through centralized shipment, route, delivery-attempt, and exception information.
- Enhance delivery-agent efficiency through structured delivery-attempt and exception recording.
- Enable business insights with reports and dashboards showing shipment status, exception trends, route activity, and delivery performance.
- Provide conversational operational assistance using **AgentForce AI and Prompt Builder**.
- Improve data security by implementing Salesforce profiles, roles, permission sets, field-level security, and sharing controls.

Gathering & Analysing User Needs

Users Involved

- **Operations Managers**

Monitor shipment operations, unresolved exceptions, route activity, and operational performance.

- **Dispatch Coordinators**

Assign shipments to routes, coordinate delivery activities, and monitor shipment milestones.

- **Delivery Agents**

Manage assigned shipments, record delivery attempts, report delivery problems, and update delivery milestones.

- **Support Team**

Review shipment and exception information and assist with operational queries.

- **Admin Team**

Configure Salesforce, manage users, security settings, automation, and operational configuration.

- **System Integrator / Developer**

Responsible for Salesforce configuration, Flow automation, Apex development, AI setup, testing, and system implementation.

Key Functional Needs

- Ability to create and manage shipment records.
- Ability to assign shipments to appropriate delivery routes.
- Ability to update shipment milestones such as:

Created → Scheduled → Dispatched → In Transit → Out for Delivery → Delivered

- Ability for delivery agents to record multiple delivery attempts.
- Ability to create and classify delivery exceptions.
- Ability to assign high-priority or critical exceptions to responsible operations users.
- Automated notifications and follow-up tasks for important operational events.
- Ability to maintain proof-of-delivery information.
- AgentForce AI integration through Prompt Builder for shipment and exception-related queries.
- Reports and dashboards to monitor shipment status, delivery attempts, route activity, exceptions, and operational performance.
- Security controls using Salesforce profiles, roles, permission sets, field-level security, and sharing rules.

Tools Used

- **Salesforce Developer Org** – For configuration, development, testing, and deployment of the SwiftShip Tracker solution.

- **Salesforce Flow Builder** – To automate shipment status updates, exception assignment, follow-up activities, and notifications.
- **Prompt Builder / AgentForce AI** – To provide conversational shipment and exception assistance.
- **Apex** – For complex operational processing where declarative Salesforce automation is insufficient.
- **Reports & Dashboards** – To provide shipment and exception monitoring.
- **Google Sheets / Docs** – For initial requirement planning, data modeling, project documentation, and system mapping.

Requirement Analysis



Shipment & Route Data Model



Delivery Attempt & Exception Management



Flows + Apex



Reports & Dashboards



AgentForce + Prompt Builder



Testing



Documentation & Submission

Identifying Key Salesforce Features & Tools Required

Salesforce Features Planned:

- **Custom Objects:**

Shipment__c, Route__c, Delivery_Attempt__c, Delivery_Exception__c, Proof_of_Delivery__c

- **Standard Objects:**

Account, Contact, Case, Task, User

These standard objects can support operational users, support activities, follow-up tasks, and user management.

- **Automations:**

Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows

These automations are planned for shipment milestone updates, exception assignment, follow-up tasks, and operational notifications.

- **Apex:**

Apex Classes can be used for complex shipment-performance processing, exception-related calculations, and operational logic that cannot be efficiently handled using declarative automation.

Batch Apex can be used where periodic processing of larger volumes of shipment or exception records is required.

- **UI:**

Lightning App Page (SwiftShip Tracker App), Lightning Record Pages, Dynamic Forms, and related list components

These provide operations users with access to shipment, route, delivery-attempt, exception, and proof-of-delivery information.

- **AgentForce AI & Prompt Builder:**

Integrated AI assistance is planned to help authorized operations users retrieve shipment information, identify unresolved exceptions, and summarize delivery-related issues using conversational queries.

- **Email Services:**

Custom Email Alerts and Email Templates can be used for:

- Critical exception notifications

- Exception assignment notifications
- Follow-up reminders
- Shipment milestone updates
- Delivery completion acknowledgments

● **Security:**

Profiles, Permission Sets, Role Hierarchy, Field-Level Security, and Sharing Rules

These controls provide appropriate access for Operations Managers, Dispatch Coordinators, Delivery Agents, Support Staff, and Administrators.

Designing Data Model and Security Model

Data Model Includes:

● **Shipment__c — Custom Object**

Stores the core shipment information.

Important information includes:

- Shipment Number
- Tracking Code
- Shipment Status
- Priority
- Origin
- Destination
- Dispatch Date
- Expected Delivery Date
- Current Milestone
- Route
- Recipient Information
- Exception Flag

The Shipment object acts as the central record for the shipment lifecycle.

● **Route__c — Custom Object**

Stores route-related information used for shipment coordination.

It can contain:

- Route Name
- Route Code
- Origin
- Destination
- Distance
- Route Status
- Assigned Delivery Agent
- Planned Start
- Planned End

A route can be associated with multiple shipments.

● **Delivery_Attempt__c — Custom Object**

Stores the history of delivery attempts associated with a shipment.

It can contain:

- Shipment
- Delivery Agent
- Attempt Date
- Attempt Number
- Attempt Status
- Remarks
- Recipient Availability
- Next Attempt Date

This object allows the system to maintain a structured history of delivery activities.

● **Delivery_Exception__c — Custom Object**

Stores problems encountered during the shipment or delivery process.

It can contain:

- Exception Number
- Shipment
- Exception Type
- Severity
- Description
- Reported Date
- Assigned To
- Resolution Status
- Resolution Notes
- Resolution Date
- Escalation Required

This object is a major component of the new Project 2 because the project focuses on **delivery exception management**.

● **Proof_of_Delivery__c — Custom Object**

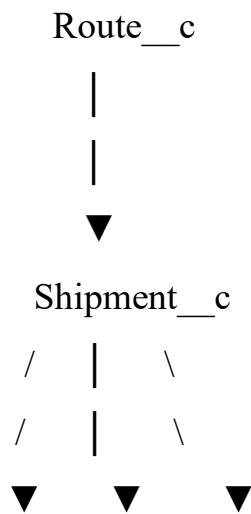
Stores information confirming successful delivery.

It can contain:

- Shipment
- Delivery Date
- Receiver Name
- Confirmation Type
- Confirmation Status
- Delivery Remarks
- Recorded By

This provides a structured record of delivery completion.

Data Relationships



Delivery_Attempt **Delivery_Exception** **Proof_of_Delivery**

The **Shipment__c** object acts as the central operational record, while the other objects maintain related route, delivery, exception, and completion information.

Security Model Design

• Role Hierarchy:

Administrator → **Operations Manager** → **Dispatch Coordinator** → **Delivery Agent**

Support Staff can be provided with appropriate access to support and operational records based on their responsibilities.

• Profiles:

Admin Profile:

Full administrative access to the SwiftShip Tracker configuration and operational data.

Operations Manager Profile:

Access to shipment, route, exception, delivery-attempt, proof-of-delivery, reports, and dashboards required for operational monitoring.

Dispatch Coordinator Profile:

Access required to assign routes, monitor shipments, and coordinate delivery activities.

Delivery Agent Profile:

Read/Edit access to assigned shipment, delivery-attempt, and relevant exception records.

Support Staff Profile:

Access to shipment and exception information required for handling operational queries and support activities.

● **Permission Sets:**

Permission Sets can be used to provide additional privileges without changing the user's primary profile.

Examples:

- **Update Shipment Milestone**
- **Manage Delivery Exceptions**
- **Access Operations Dashboard**
- **Manage Route Assignments**
- **Use SwiftShip AI Assistant**

● **Record-Level Security:**

Sharing Rules can be applied to control access to shipment and operational records.

For example:

- Operations Managers can access relevant shipment and exception records.
- Dispatch Coordinators can access shipments associated with their operational responsibilities.
- Delivery Agents can access assigned shipment and delivery records.
- Support Staff can access records required for issue resolution.

● **Field-Level Security:**

Sensitive or operationally restricted fields can be controlled using Field-Level Security.

Examples include:

- Recipient Contact
- Recipient Phone
- Internal Resolution Notes
- Exception Escalation Information
- Operational Remarks

Access should be limited according to the user's role and responsibilities.

Phase 2: Salesforce Development – Backend & Configurations

This phase focuses on developing the core Salesforce functionality of **Swift Ship Tracker**.

Custom objects are created for **Shipment, Route, Delivery Attempt, Delivery Exception, and Proof of Delivery**, with appropriate relationships between the records.

Automation is developed using Salesforce Flows to manage shipment milestones, exception assignments, follow-up activities, and notifications.

Apex can be introduced for complex operational processing, while AgentForce AI and Prompt Builder provide conversational assistance for authorized users.

Reports and dashboards are configured to provide visibility into shipment status, route activity, delivery attempts, unresolved exceptions, and operational performance.

Milestone 1 – Salesforce Account

Introduction:

Salesforce provides the platform on which the Swift Ship Tracker solution is configured and developed.

For this project, the Salesforce environment is used to create the required custom objects, relationships, automation, security controls, Lightning interfaces, reports, dashboards, and AI-assisted functionality.

The Salesforce Developer environment provides the foundation for building and testing the SwiftShip Tracker application before the solution is considered for further deployment.

What Is Salesforce?

Salesforce is a cloud-based CRM platform that provides tools for managing data, business processes, automation, reporting, security, and application development.

In the Swift Ship Tracker project, Salesforce is used as the central platform for managing **shipment operations, route information, delivery activities, delivery exceptions, and proof of delivery.**

The project uses Salesforce capabilities such as:

- Custom Objects
- Lightning App
- Lightning Record Pages
- Salesforce Flow
- Apex
- Reports and Dashboards
- AgentForce AI
- Prompt Builder
- Profiles and Permission Sets
- Role Hierarchy
- Field-Level Security
- Sharing Rules

Creating Developer Account

Creating a developer org in salesforce.

1. <https://developer.salesforce.com/signup>
2. On the sign-up form, enter the following details:

salesforce.com/products/free-trial/developer/

**Build for the Agentic Enterprise.
Sign up for your Developer Edition.**

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name: Dhaarani ✓ Last name: V ✓

Job title: Salesforce Developer ✓ Company: The SmartBridge ✓

Country/region: India ✓ Work email: dhaaranivenugopal16@g... ✓

☒ I agree to the [Main Services Agreement – Developer Services, Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

Sign me up

1) First name & Last name

2) Email

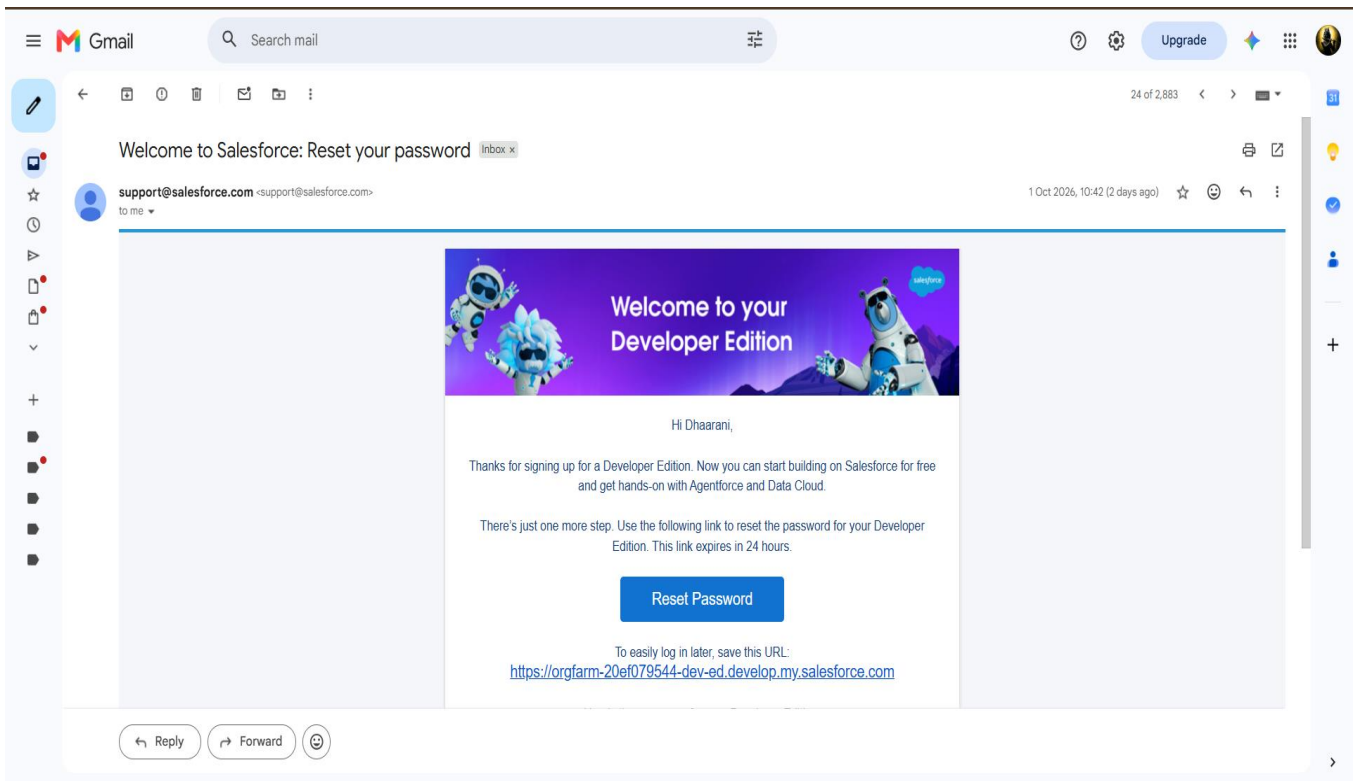
3) Job Title: Developer

4) Company: College Name

5) Country: India

Click on sign me up after filling these.

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30mins and sometimes 2 hours.



2. Click on Verify Account

3. Give a password and answer a security question and click on change password.

Make sure to include at least:

- ✓ 8 characters
- ✓ 1 letter
- ✓ 1 number

* New Password

..... Good

* Confirm New Password

..... Match

Security Question

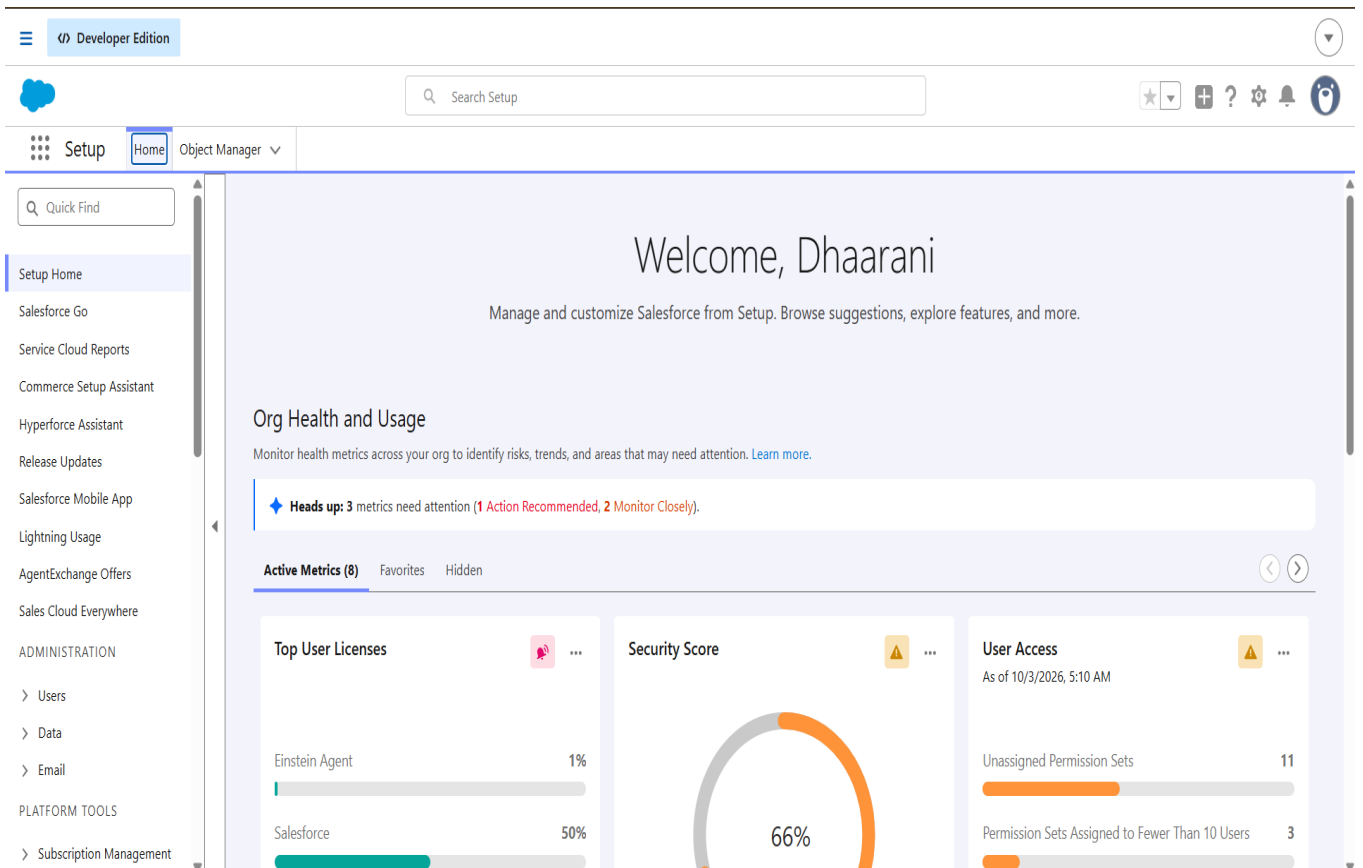
Thiruvarur

* Answer

.....

Change Password

4. Then you will redirect to your salesforce setup page.



Milestone 2- Objects Creation

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Parcel

5. Enter Plural label name as Parcels
6. Enter Record Name as Parcel Name
7. Select Data Type as Text: Parcel Name
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New

SETUP

New Custom Object

Optional Features

☒ Allow Reports

☒ Allow Activities

☒ Track Field History

☐ Allow in Chatter Groups

☐ Enable Licensing [i](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#)

☒ Allow Sharing

☒ Allow Bulk API Access

☒ Allow Streaming API Access

Deployment Status

[What is this?](#)

☐ In Development

☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more](#)

☒ Allow Search

Object Creation Options (Available only when custom object is first created)

☒ Add Notes and Attachments related list to default page layout

☐ Launch New Custom Tab Wizard after saving this custom object

Save

Save & New

Cancel



Q

Search Setup





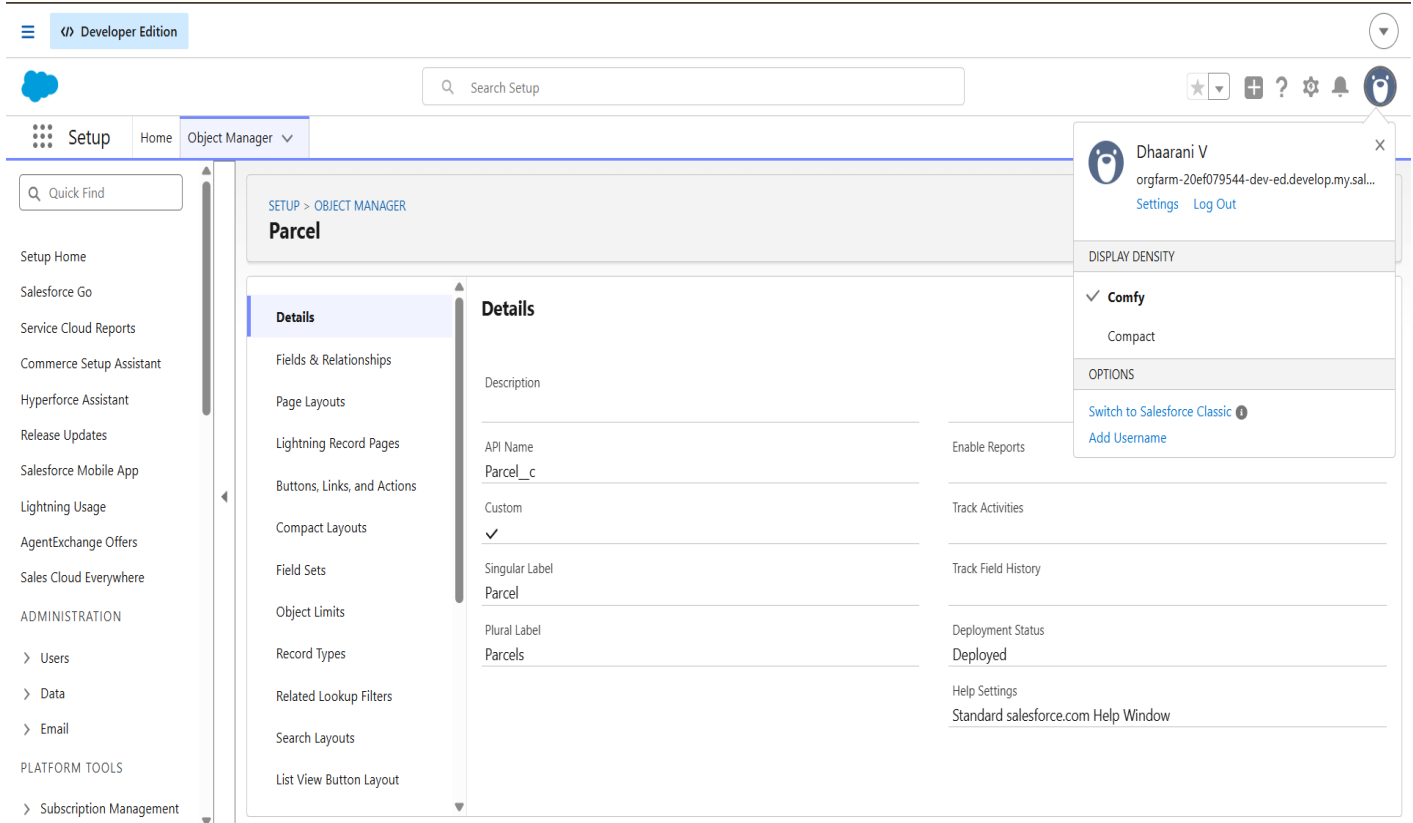












Developer Edition

Search Setup

Setup Home Object Manager

Quick Find

SETUP > OBJECT MANAGER

Parcel

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Details

Description

API Name
Parcel__c

Custom
✓

Singular Label
Parcel

Plural Label
Parcels

Enable Reports

Track Activities

Track Field History

Deployment Status
Deployed

Help Settings
Standard salesforce.com Help Window

Dhaarani V
orgfarm-20ef079544-dev-ed.develop.my.sal...
Settings Log Out

DISPLAY DENSITY

✓ Comfy

Compact

OPTIONS

Switch to Salesforce Classic ⓘ

Add Username

Creating Delivery Object

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Delivery
5. Enter Plural label name as Deliveries
6. Enter Record Name as DeliveryName
7. Select Data Type as Text: DeliveryName
8. Select Allow reports.
9. Select Allow search.

11. Click on Save and New

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Delivery

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Details

Edit

Delete

Description

API Name

Delivery_c

Custom

✓

Singular Label

Delivery

Plural Label

Deliveries

Enable Reports

✓

Track Activities

✓

Track Field History

✓

Deployment Status

Deployed

Help Settings

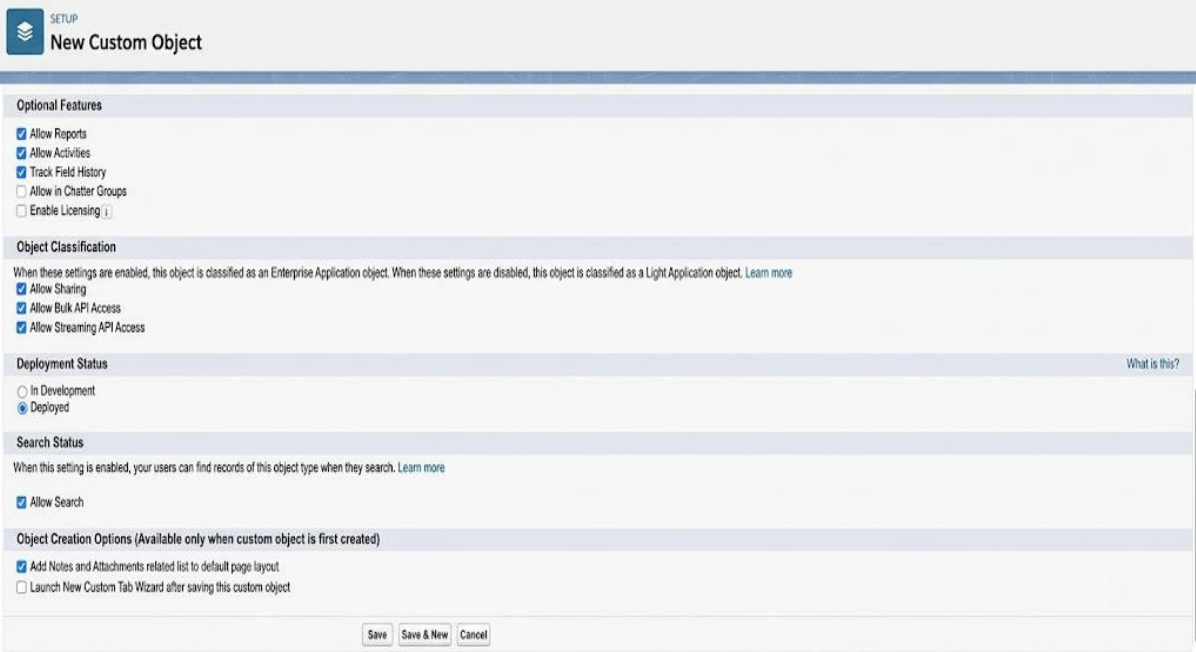
Standard salesforce.com Help Window

Description

Creating Sender Object

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Sender
5. Enter Plural label name as Sender's
6. Enter Record Name as SenderName
7. Select Data Type as Text: SenderName
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New



The screenshot shows the 'New Custom Object' setup page in Salesforce. The page is titled 'New Custom Object' and includes a 'SETUP' icon. It is divided into several sections with checkboxes for configuration:

- Optional Features:**
 - ☒ Allow Reports
 - ☒ Allow Activities
 - ☒ Track Field History
 - ☐ Allow in Chatter Groups
 - ☐ Enable Licensing
- Object Classification:**

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#)

 - ☒ Allow Sharing
 - ☒ Allow Bulk API Access
 - ☒ Allow Streaming API Access
- Deployment Status:**

[What is this?](#)

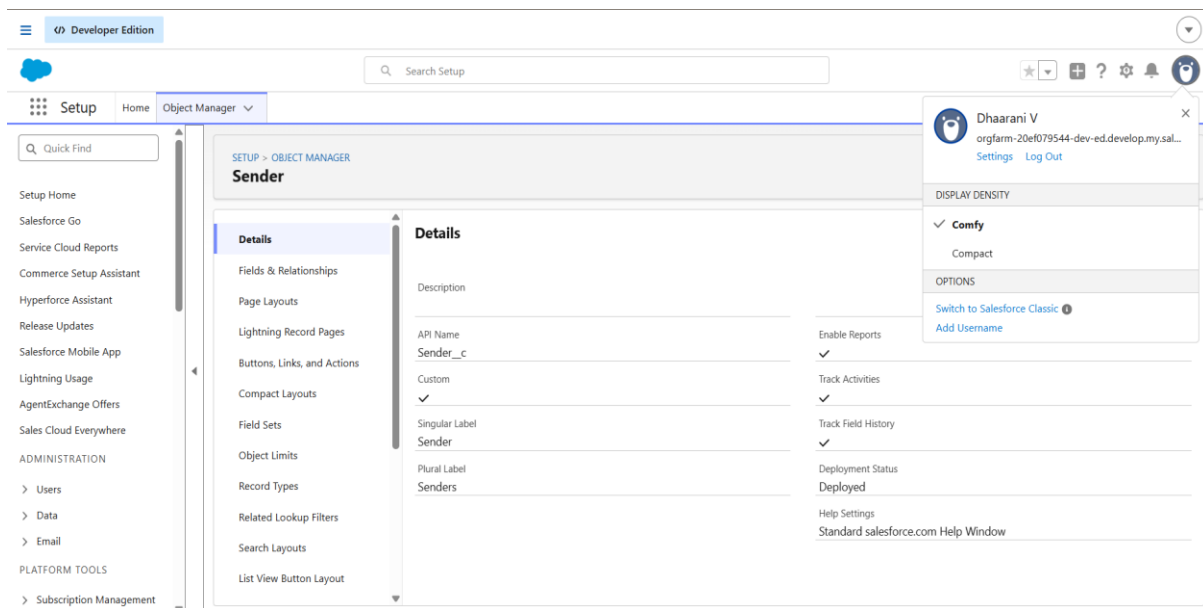
 - ☐ In Development
 - ☒ Deployed
- Search Status:**

When this setting is enabled, your users can find records of this object type when they search. [Learn more](#)

 - ☒ Allow Search
- Object Creation Options (Available only when custom object is first created):**
 - ☒ Add Notes and Attachments related list to default page layout
 - ☐ Launch New Custom Tab Wizard after saving this custom object

At the bottom of the form are three buttons: 'Save', 'Save & New', and 'Cancel'.

The footer of the page includes the Salesforce logo, a search bar labeled 'Search Setup', and a row of utility icons (star, plus, home, question mark, gear, bell, and user profile).



Creating Receiver Object

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Receiver
5. Enter Plural label name as Receiver's
6. Enter Record Name as ReceiverName
7. Select Data Type as Text: ReceiverName
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New

SETUP
New Custom Object

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [?](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status [What is this?](#)

- ☐ In Development
- ☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more](#)

- ☒ Allow Search

Object Creation Options (Available only when custom object is first created)

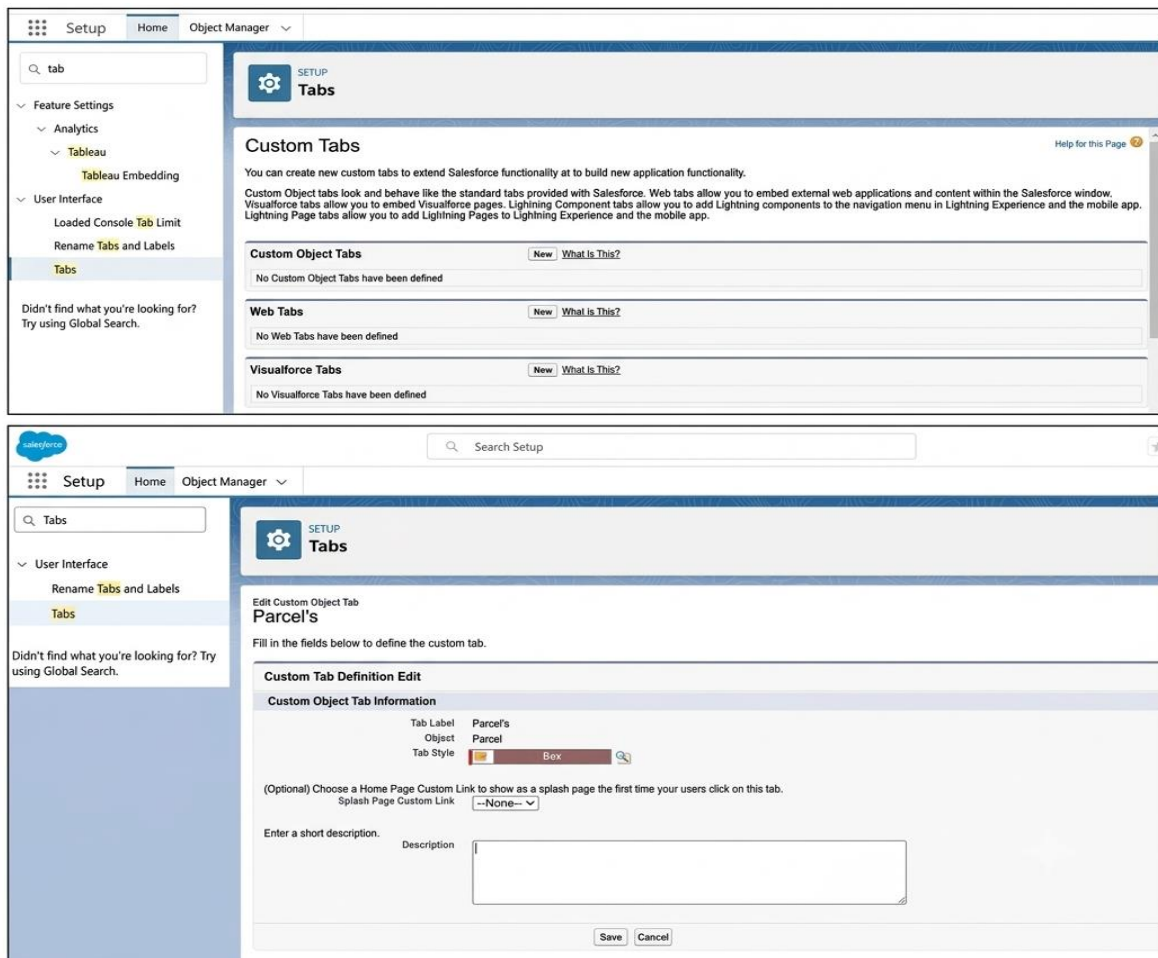
- ☒ Add Notes and Attachments related list to default page layout
- ☐ Launch New Custom Tab Wizard after saving this custom object

[Save](#) [Save & New](#) [Cancel](#)

Milestone 3 – Tabs

Creating a tab for Parcel Object

1. Go to the setup page → type Tabs in Quick Find bar
2. Click on tabs
3. Click on New (under custom object tab).
4. Select Object (Parcel) >> Select the tab style
5. Click on Next >> (Add to profiles page) keep it as default >> Click on Next (Add to Custom App) uncheck the include tab.
6. Make sure that the Append tab to the user's existing personal customizations is checked.
6. Click save



More Tab Creation

1. Go to Setup → type Tabs in the Quick Find box.
2. Click Tabs under the User Interface section.
3. Click New under Custom Object Tabs.
4. Select the object (Delivery, Sender, or Receiver) → choose a Tab Style.
5. Click Next → keep the Add to Profiles settings as default → click Next.
6. On the Add to Custom Apps page, uncheck “Include Tab.”
7. Ensure Append tab to the user’s existing personal customizations is checked.
8. Click Save and repeat the steps for each object (Delivery, Sender, Receiver).

Developer Edition

Search Setup

Setup Home Object Manager

tab

Feature Settings

Analytics

Tableau

Tableau Embedding

Tableau UAF Claims Definition

User Interface

Console Settings

Loaded Console Tab Limit

Rename Tabs and Labels

Tabs

Didn't find what you're looking for? Try using Global Search.

SETUP Tabs

Custom Tabs

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Experience and the mobile app.

Custom Object Tabs

New What Is This?

Action	Label	Tab Style	Description
Edit Del	Deliveries	Box	
Edit Del	Parcels	Caduceus	
Edit Del	Receivers	Big top	
Edit Del	Senders	Apple	

Web Tabs

New What Is This?

No Web Tabs have been defined

Visualforce Tabs

New What Is This?

Dhaarani V

orgfarm-20ef079544-dev-ed.develop.my.salesforce.com

Settings Log Out

DISPLAY DENSITY

Comfy

Compact

OPTIONS

Switch to Salesforce Classic

Add Username

Milestone 4- Fields & Relationships

Create fields and Relationships for the custom objects created in the previous milestone.

Creating Fields and Relationships for Parcel Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Parcel object.

Field 1: Parcel ID

- Go to Fields & Relationships → click New

Enter:

- Data type== auto number→P-{000}

SETUP > OBJECT MANAGER

Parcel

Details

Fields & Relationships
9 Items, Sorted by Field Label

Quick Find:

[New](#) [Deleted Fields](#) [Field Dependencies](#) [Set History Tracking](#)

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		

- Click Next → Next → Save & New.

3. Follow step 1 and 2 to create the remaining fields(Status, Weight, Estimated Delivery, Sender) of the Parcel object mentioned in the table below.)

1.Parcel Object

API Name:Parcel_c

Purpose :Store parcel information

Field Name	API Name	Data Type	Description
Parcel ID	Parcel_ID__c	Auto number	Unique ID for each parcel
Status	Status__c	Picklist	[Booked, In Transit, Out for Delivery, Delivered]
Weight	Weight__c	Number	Weight of Parcel
Estimated Delivery	Estimated_Delivery_Date__c	Date	Expected Delivery Date
Sender	Sender__c	Lookup Relationship	Gets Details from Sender

Creating Fields And Relationships For Sender Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Sender object.

Field 1: Sender Address

- Go to Fields & Relationships → click New

The screenshot shows the Salesforce Setup interface. At the top, there's a 'Developer Edition' label and a search bar. Below the navigation bar, the 'Object Manager' tab is selected. On the left sidebar, 'Fields & Relationships' is highlighted under the 'Receiver' object. The main content area shows a table with the following data:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		

At the top of the table, it says '9 Items, Sorted by Field Label'. There are also buttons for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking'.

- Choose Data Type: Geolocation → click Next.

Field Label: Sender Address

Click Next → Next → Save & New.

The screenshot shows the Salesforce Setup interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The left sidebar shows the 'Delivery' object selected. The main content area displays the 'New Custom Field' wizard for the 'Delivery' object. The wizard is at Step 2 of 4, 'Enter the details'. The 'Field Label' is 'Current Location', the 'Field Name' is 'Current_Location', and the 'Decimal Places' are set to 15. The 'Latitude and Longitude Display Notation' is set to 'Degrees, Minutes, Seconds'. The 'Required' checkbox is unchecked.

3. Follow step 1 and 2 to create the remaining fields (Sender Contact, Sender Email) of the Sender object mentioned in the table below.

Creating Fields And Relationships For Receiver Object

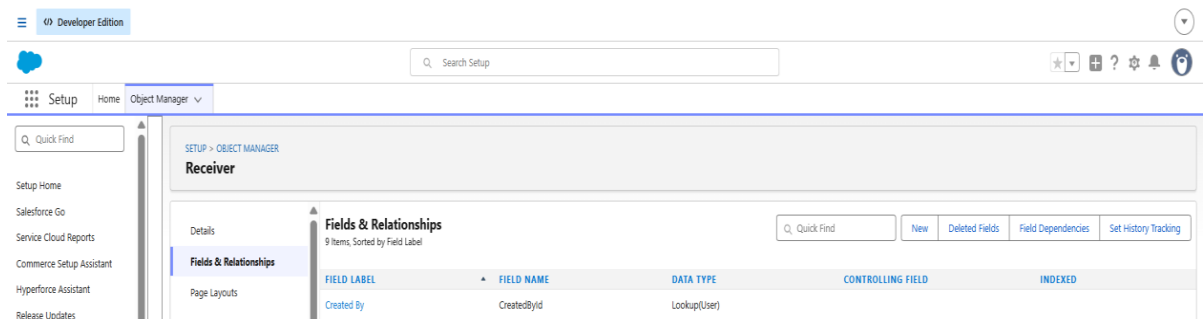
Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Receiver object.

Field 1: Receiver Address

Field Name	API Name	Data Type	Description
Sender Address	Sender_Adress__c	Geolocation	Location in terms of longitude and latitude.
Sender Contact	Sende_Contact__c	Phone	To contact the sender
Sender Email	Sender_Email__c	Email	To send email Notification.

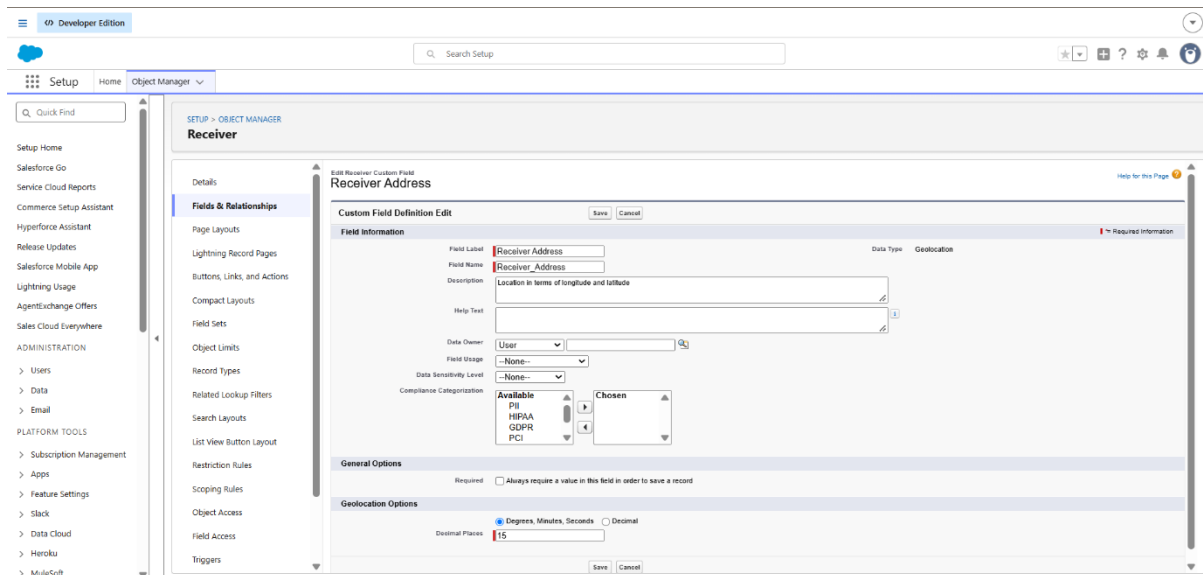
- Go to Fields & Relationships → click New



- Choose Data Type: Geolocation → click Next.

Field Label: Receiver's Address

Click Next → Next → Save & New.



- 3. Follow step 1 and 2 to create the remaining fields (Receiver Contact, Receiver Email, Sender, Parcel) of the Receiver object mentioned in the table below.

Field Name	API Name	Data Type	Description
Receiver Address	Receiver_Adress__c	Geolocation	Location in terms of longitude and latitude.
Receiver Contact	Receiver_Contact__c	Phone	To contact the Receiver
Receiver Email	Receiver_Email__c	Email	To send email Notification.
Sender	Sender__c	Lookup Relationship	Gets Details from Sender
Parcel	Parcel__c	Lookup Relationship	Gets Details from Parcel

Milestone 5 - Prompt Builder

Designing Prompts for Agent Building.

In Salesforce, go to Setup.

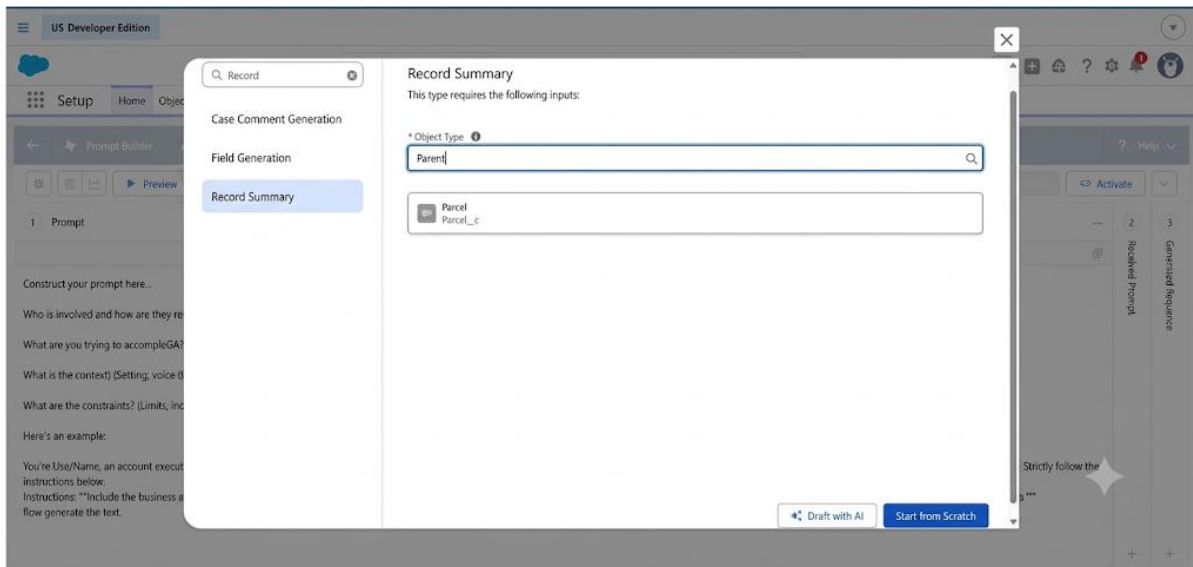
1. In the Quick Find box, type Prompt Templates (or in Agent Console, go to Agent Builder →

Prompt Templates).

2. Click New Prompt Template.-- Record summary template

The screenshot displays the Salesforce Prompt Builder interface. The left sidebar contains the navigation menu with 'Prompt Builder' selected under 'Einstein Generative AI'. The main content area is titled 'Prompt Builder' and includes a 'New Prompt Template' button. Below this, there are three cards: 'Get to Know Prompt Builder' (with a 'Watch Video' button), 'Build Your Knowledge' (with a 'Learn' button), and 'Take a Tour of Prompt Builder' (with a 'Start Tour' button). At the bottom, there is a table titled 'Prompt Templates' showing a list of templates.

Name	Template Type	Category	Status	Date Modified
Extract Branding Info	Global Standard	Standard	● Active	
Deflection Score (Defence)	Agentforce Scorer Measurement	Standard	● Active	
Cancel Assets	Global Standard	Standard	● Active	
Guided Search - Default Instructions	Guided Search Instructions	Standard-Overridable	● Active	
Revenue Reconciliation Analysis	Global Standard	Standard	● Active	
Create Quote with Pricing	Global Standard	Standard	● Active	



5. Enter the Prompt Template Content

In the Prompt Template Text area, paste the following exactly:

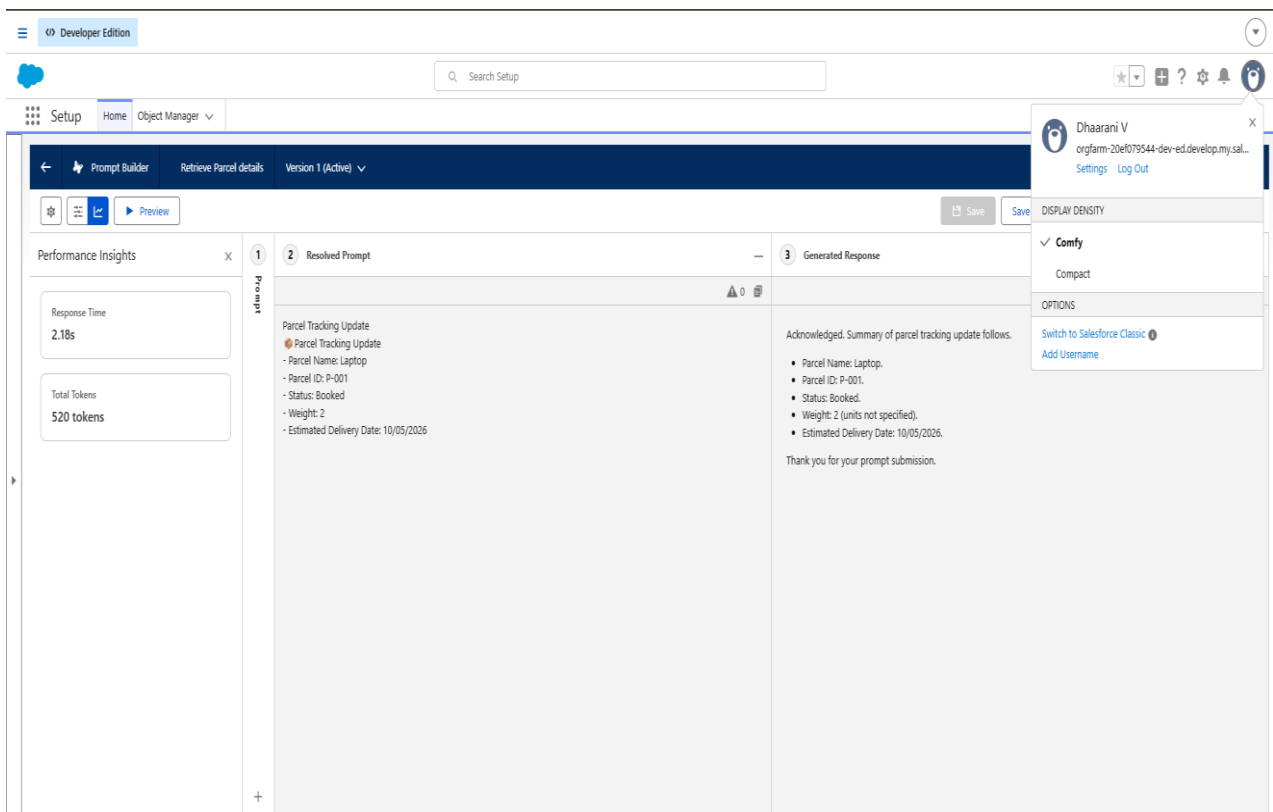
Retrieve details from Parcel c where Parcel_ID__c matches user input. Return response in this

format:

Parcel Tracking Update

Parcel Tracking Update

- Parcel Name: {!\$Input:Parcel__c.Name}
- Parcel ID: {!\$Input:Parcel__c.Parcel_ID__c}
- Status: {!\$Input:Parcel__c.Status__c}
- Weight: {!\$Input:Parcel__c.Weight__c}
- Estimated Delivery Date: {!\$Input:Parcel__c.Estimated_Delivery_Date__c}



6. Add Resource Mapping

7. Scroll down to the Resources / Insert Resources section.

- Click Insert Resource → choose Object.

- Select Parcel__c.

- Add the following fields:

- Name

- Parcel_ID__c

- Status__c

- Weight__c

- Estimated_Delivery_Date__c

8. Click Insert Resource (or Add) for each one.

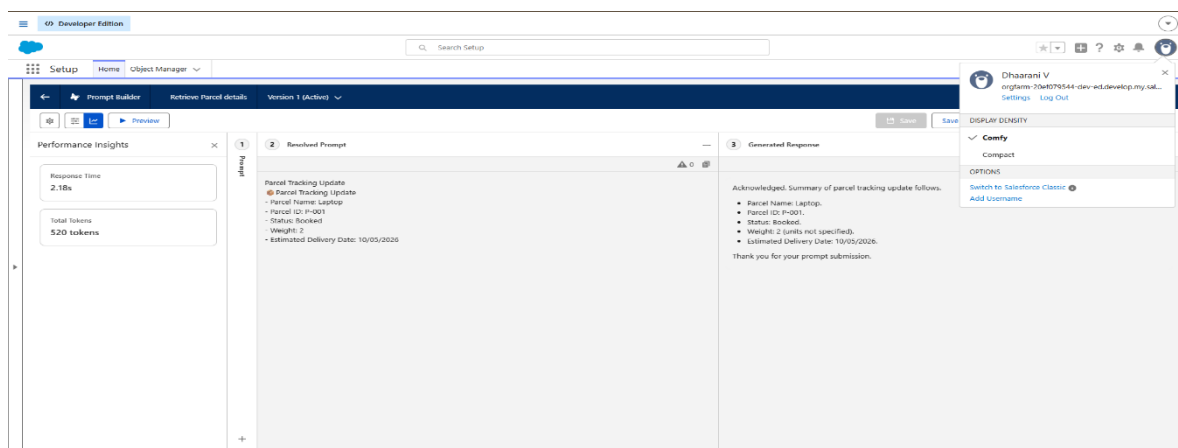
9. Click Save.

10. Then click Activate to make this Prompt Template available for your agent or flow.

Testing the Prompt

Test the Prompt:

1. Open your Agent preview.
2. Type: “Track parcel P-12345”.
3. It should return details from the Parcel record.

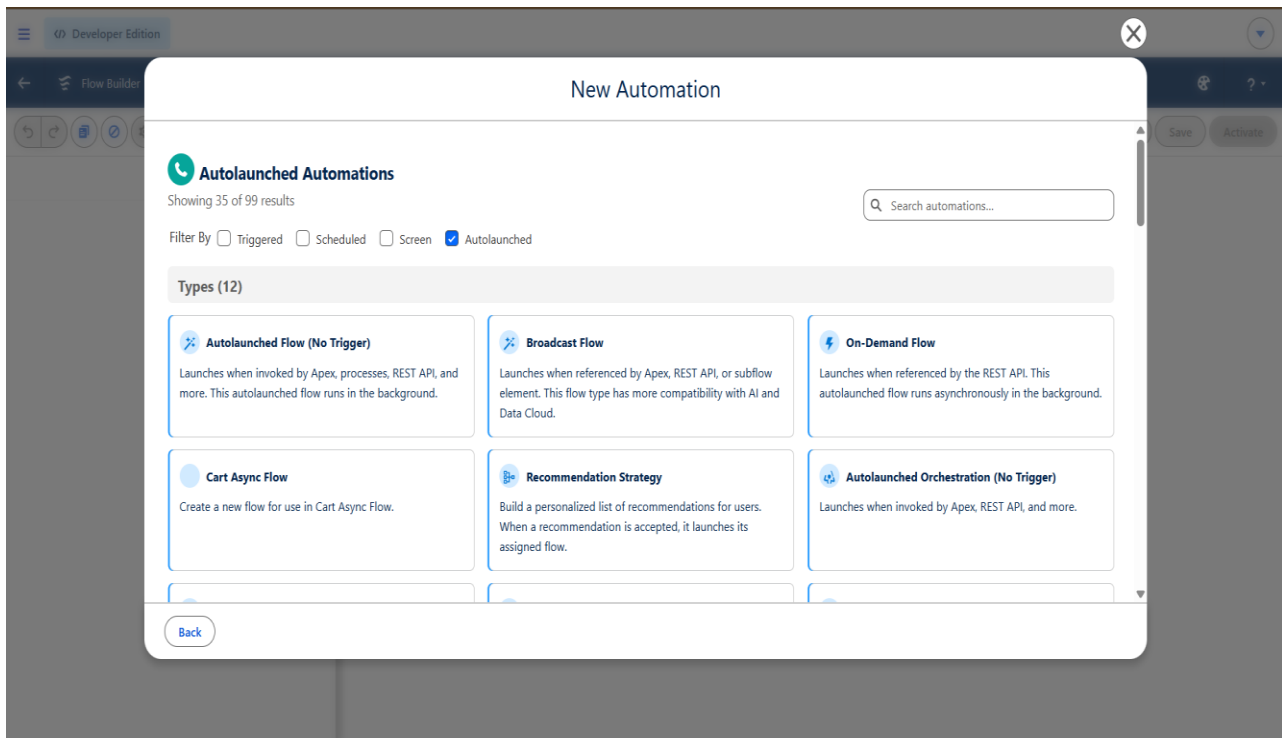


Milestone 6 - Flows

Create flows for capturing the response as shown.

Go to Setup → search Flows → click New Flow.

1. Choose Auto-Launched Flow (No Trigger) → click Create



3. Add Input Variables (to allow Agent to pass values):

- Click On New Recourse → Variable → Available for Input.

New Resource

*Resource Type
Variable

*API Name ⓘ
Ids

Description

*Data Type
Text ☐ Allow multiple values (collection) ⓘ

Default Value
Enter value or search resources...

Availability Outside the Flow
☒ Available for input
☐ Available for output

Cancel Done

- Ids → Data Type: Text → Available for Input.
- Output → Data Type: Text → Available for Output.

New Resource

*Resource Type
Variable

*API Name ⓘ
output

Description

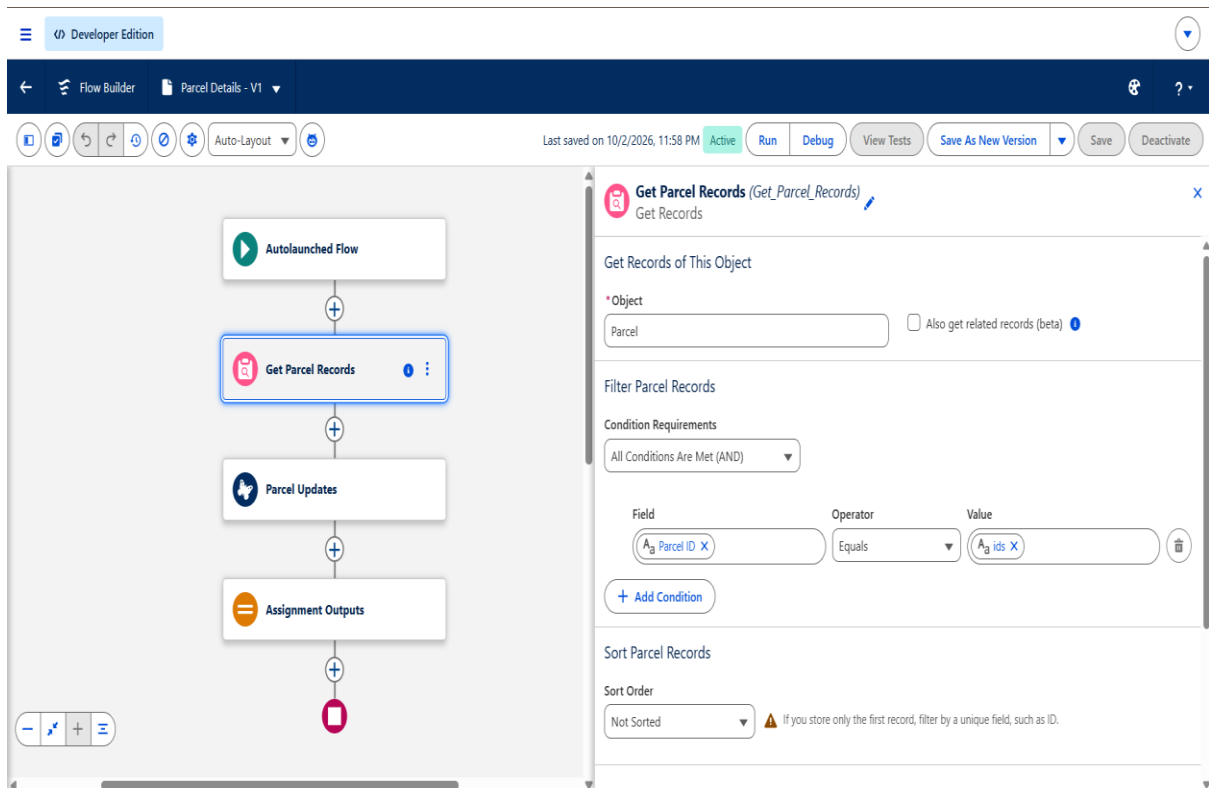
*Data Type
Text ☐ Allow multiple values (collection) ⓘ

Default Value
Enter value or search resources...

Availability Outside the Flow
☐ Available for input
☒ Available for output

Cancel Done

4. Click on the Add icon to add Element: Get Records→ Label: Get Parcel Records

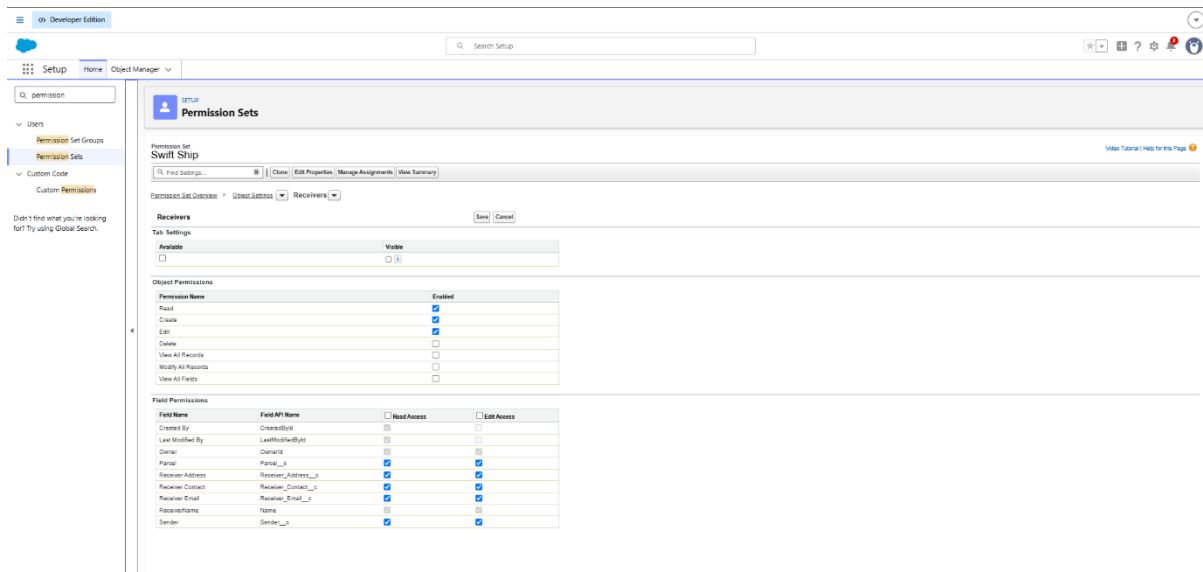


- Object: Parcel
- Field = Parcel_ID__c
- Operator = Equals
- Value = ids

5. Element : Action

1. Find Retrieve parcel details
2. Click Retrieve parcel details

- Label: Parcel Updates.
- Object To Summarize : Parcel from get Parcel records.

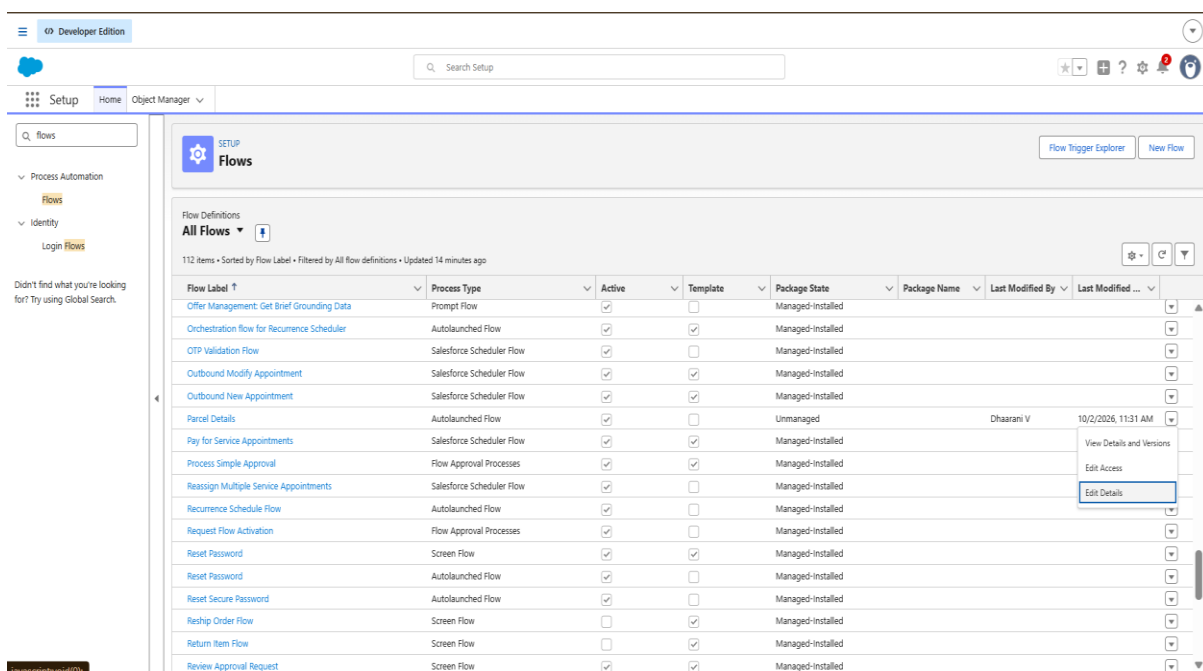


7. Connect Start → Get Records → Action → Assignment → End.

8. Save Flow → Name: Parcel Details.

9. Activate Flow.

10. Once the flow is saved go back to the flows, Click on the drop down near the flow that is just created, click on edit access.



Phase 3: UI/UX Development & Customization

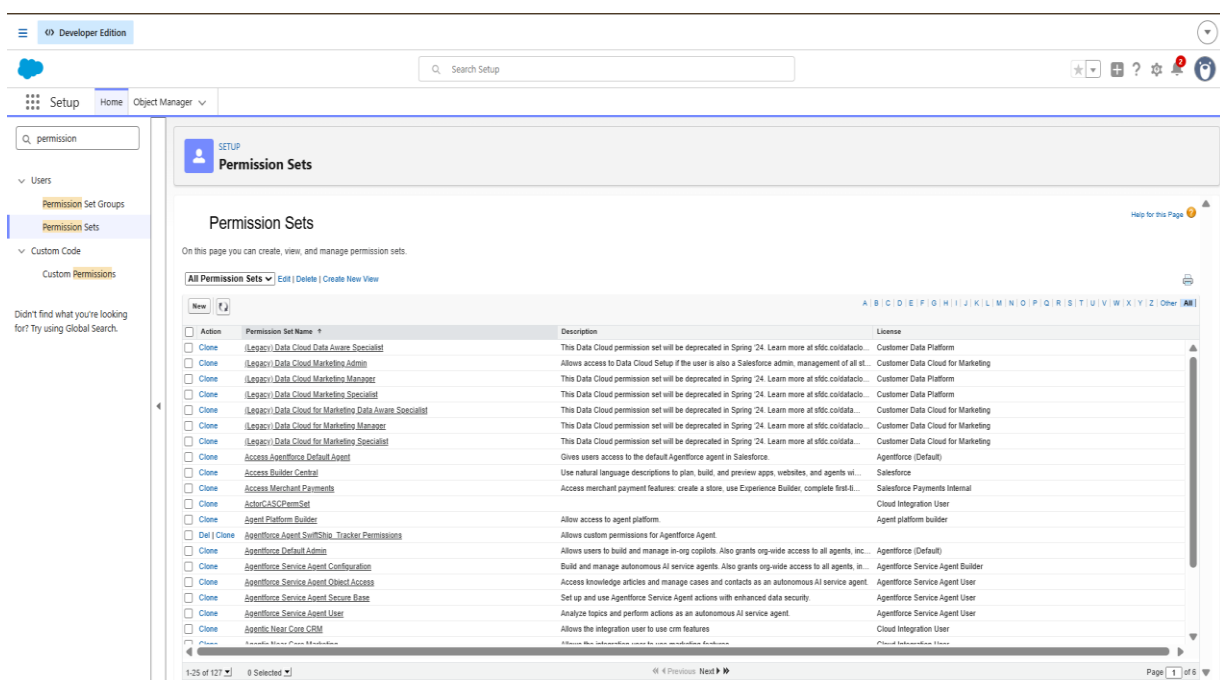
To enable agents to efficiently manage parcel operations and ensure secure access control using

permission sets and AI-driven agent experiences built with AgentForce.

Milestone - 1 Creating Permission Sets

In the Quick Find box (left side), type Permission Sets.

1. Click Permission Sets under Users.



The screenshot shows the Salesforce Setup interface. On the left, the 'Quick Find' box contains the text 'permission'. Below it, the navigation menu is expanded to 'Users', and 'Permission Sets' is selected. The main content area is titled 'Permission Sets' and includes a sub-header 'On this page you can create, view, and manage permission sets.' Below this, there is a table of existing permission sets. The table has columns for 'Action', 'Permission Set Name', 'Description', and 'License'. The first few rows show permission sets like 'Legacy Data Cloud Data Aware Specialist' and 'Legacy Data Cloud Marketing Admin'. The bottom of the page shows '1-25 of 127' and '0 Selected'.

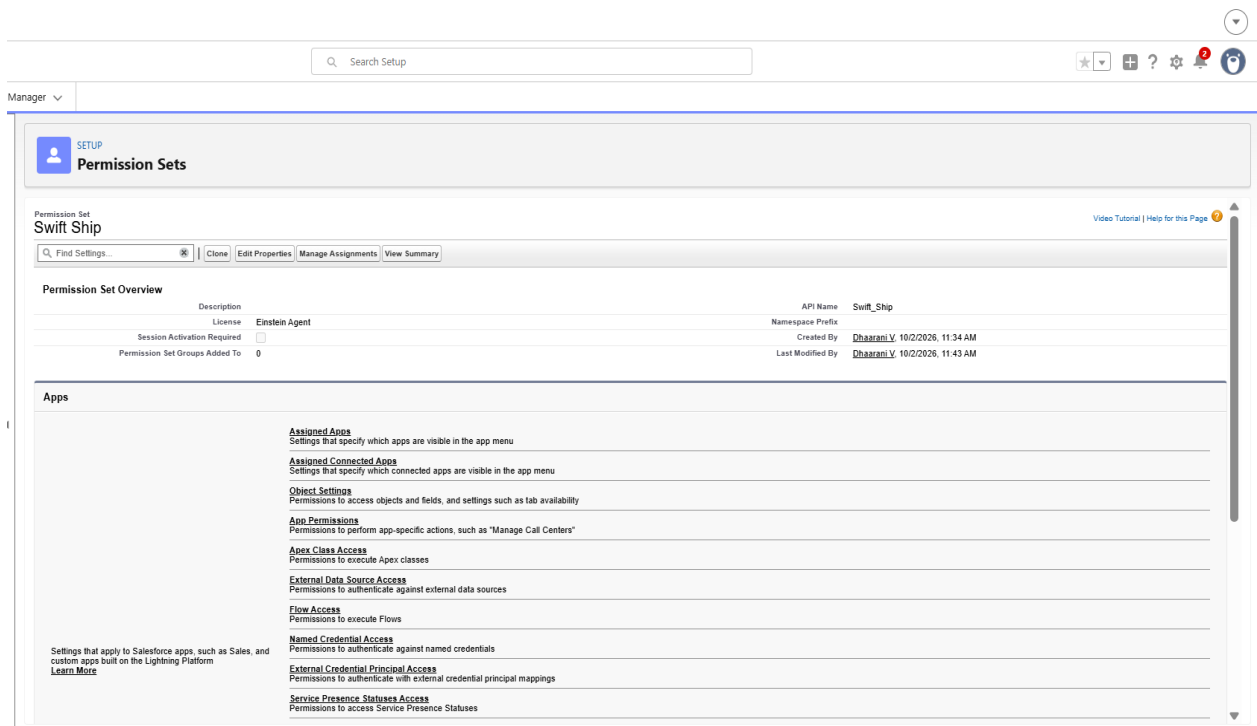
Action	Permission Set Name	Description	License
☐ Clone	Legacy Data Cloud Data Aware Specialist	This Data Cloud permission set will be deprecated in Spring '24. Learn more at sfdc.co/datacloud...	Customer Data Platform
☐ Clone	Legacy Data Cloud Marketing Admin	Allows access to Data Cloud Setup if the user is also a Salesforce admin, management of all st...	Customer Data Cloud for Marketing
☐ Clone	Legacy Data Cloud Marketing Manager	This Data Cloud permission set will be deprecated in Spring '24. Learn more at sfdc.co/datacloud...	Customer Data Platform
☐ Clone	Legacy Data Cloud Marketing Specialist	This Data Cloud permission set will be deprecated in Spring '24. Learn more at sfdc.co/datacloud...	Customer Data Platform
☐ Clone	Legacy Data Cloud for Marketing Data Aware Specialist	This Data Cloud permission set will be deprecated in Spring '24. Learn more at sfdc.co/datacloud...	Customer Data Cloud for Marketing
☐ Clone	Legacy Data Cloud for Marketing Manager	This Data Cloud permission set will be deprecated in Spring '24. Learn more at sfdc.co/datacloud...	Customer Data Cloud for Marketing
☐ Clone	Legacy Data Cloud for Marketing Specialist	This Data Cloud permission set will be deprecated in Spring '24. Learn more at sfdc.co/datacloud...	Customer Data Cloud for Marketing
☐ Clone	Access Agentforce Default Agent	Gives users access to the default Agentforce agent in Salesforce	Agentforce (Default)
☐ Clone	Access Merchant Central	Use natural language descriptions to plan, build, and preview apps, websites, and agents vi...	Salesforce
☐ Clone	Access Merchant Payments	Access merchant payment features: create a store, use Experience Builder, complete first-li...	Salesforce Payments Internal
☐ Clone	Access CASC/PharmSet	Allow access to agent platform.	Cloud Integration User
☐ Clone	Agent Platform Builder	Allows users to build and manage in-org copilots. Also grants org-wide access to all agents, inc...	Agent platform builder
☐ Del Clone	Agentforce Agent SwiftShip Tracker Permissions	Allows custom permissions for Agentforce Agent.	Agentforce (Default)
☐ Clone	Agentforce Default Admin	Build and manage autonomous AI service agents. Also grants org-wide access to all agents, in...	Agentforce Service Agent Builder
☐ Clone	Agentforce Service Agent Configuration	Access knowledge articles and manage cases and contacts as an autonomous AI service agent.	Agentforce Service Agent User
☐ Clone	Agentforce Service Agent Object Access	Set up and use Agentforce Service Agent actions with enhanced data security.	Agentforce Service Agent User
☐ Clone	Agentforce Service Agent Secure Base	Analyze topics and perform actions as an autonomous AI service agent.	Agentforce Service Agent User
☐ Clone	Agentforce Service Agent User	Allows the integration user to use csm features	Cloud Integration User
☐ Clone	Agentforce Near Core CRM	Access the information used by core modules in Salesforce	Cloud Integration User

1. Click the New button.

2. Enter the following details:

- Label: Swift Ship
- API Name: auto-fills
- License: Choose Einstein Agent.

3. Click Save.



1. Inside the Permission Set page → under Apps → Object Settings, click Object Settings.

1. Search and click each object one by one

2. Click Edit.

Enable the following:

✓ Read

✓ Create

✓ Edit

(Optional: Delete, if you want agents to remove records)

Under Field Permissions, ensure at least Visible is checked for all key fields.

Click Save after each object.

Parcel__c

Q Search Setup

☆

+

?

⚙

🔔

👤

Manager ▾

SETUP

Permission Sets

Permission Set

Swift Ship

Video Tutorial | Help for this Page

Q Find Settings...

Clone

Edit Properties

Manage Assignments

View Summary

Permission Set Overview

Description

License

Einstein Agent

Session Activation Required

Permission Set Groups Added To

0

API Name

Swift_Ship

Namespace Prefix

Created By

Dhaaram V

10/2/2026, 11:34 AM

Last Modified By

Dhaaram V

10/2/2026, 11:43 AM

Apps

Assigned Apps

Settings that specify which apps are visible in the app menu

Assigned Connected Apps

Settings that specify which connected apps are visible in the app menu

Object Settings

Permissions to access objects and fields, and settings such as tab availability

App Permissions

Permissions to perform app-specific actions, such as "Manage Call Centers"

Apex Class Access

Permissions to execute Apex classes

External Data Source Access

Permissions to authenticate against external data sources

Parcel__c

Tab Settings

Available

Visible

☐

☐

Object Permissions

Permission Name

Enabled

Read

☒

Create

☒

Edit

☒

Delete

☐

View All Records

☐

Modify All Records

☐

View All Fields

☐

Field Permissions

Field Name

Field API Name

☐ Read Access

☐ Edit Access

Created By

CreatedBy

☒

☐

Last Modified By

LastModifiedBy

☒

☐

Owner

OwnerId

☒

☐

Sender Address

Sender_Address__c

☒

☒

Sender Contact

Sender_Contact__c

☒

☒

Sender Email

Sender_Email__c

☒

☒

SenderName

Name

☒

☐

Delivery__c

Manager

Search Setup

Setup

Permission Sets

Permission Set

Swift Ship

Find Settings...CloseEdit PropertiesManage AssignmentsView Summary

Permission Set OverviewObject SettingsDeliveries

Deliveries

SaveCancel

Tab Settings

AvailableVisible

Object Permissions

Permission NameEnabled

ReadCreateEditDeleteView All RecordsModify All RecordsView All Fields

Field Permissions

Field NameField API NameRead AccessEdit Access

Created ByCreated_ByCurrent LocationCurrent_Location_cDeliveryNameNameEstimated DeliveryEstimated_Delivery_cLast Modified ByLastModifiedByIdOwnerOwnerIdParcelParcel_cSenderSender_c

Sender__c

Developer Edition

Search Setup

Setup

Home

Object Manager

permission

Users

Permission Set Groups

Permission Sets

Custom Code

Custom Permissions

Didn't find what you're looking for? Try using Global Search.

Setup

Permission Sets

Permission Set

Swift Ship

Find Settings...CloseEdit PropertiesManage AssignmentsView Summary

Permission Set OverviewObject SettingsSenders

Senders

SaveCancel

Tab Settings

AvailableVisible

Object Permissions

Permission NameEnabled

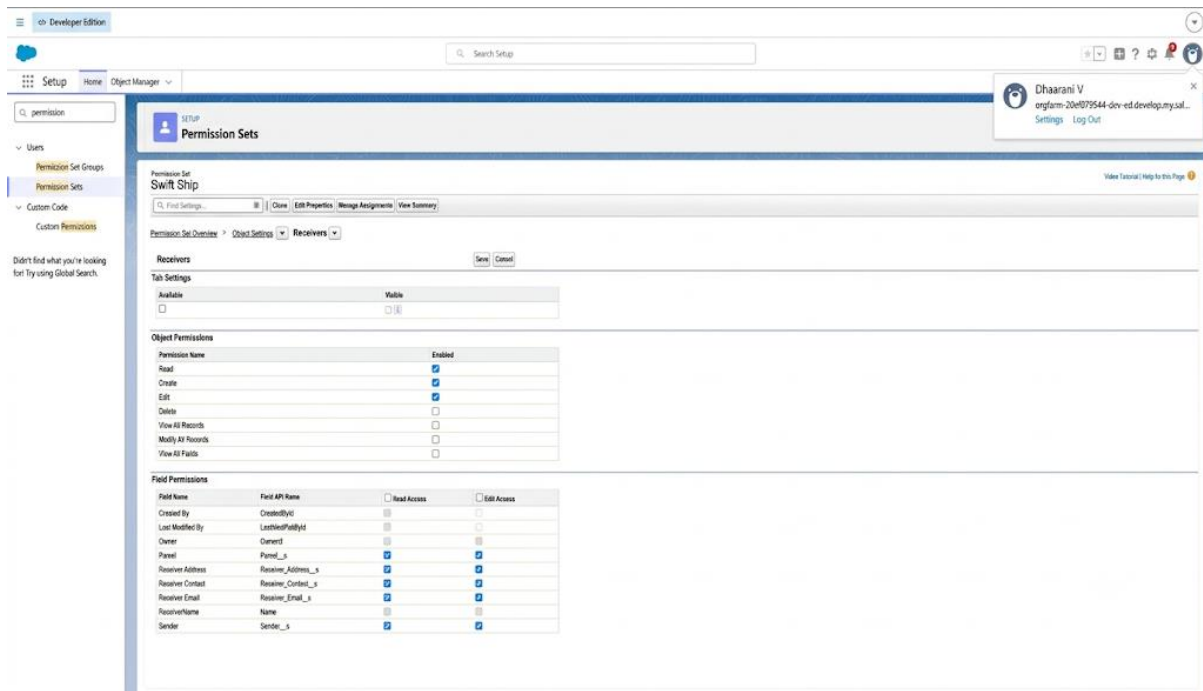
ReadCreateEditDeleteView All RecordsModify All RecordsView All Fields

Field Permissions

Field NameField API NameRead AccessEdit Access

Created ByCreated_ByLast Modified ByLastModifiedByIdOwnerOwnerIdSender AddressSender_Address_cSender ContactSender_Contact_cSender EmailSender_Email_cSender PhoneName

Receiver__c



Milestone - 2 Assigning Permissions to your Agent user

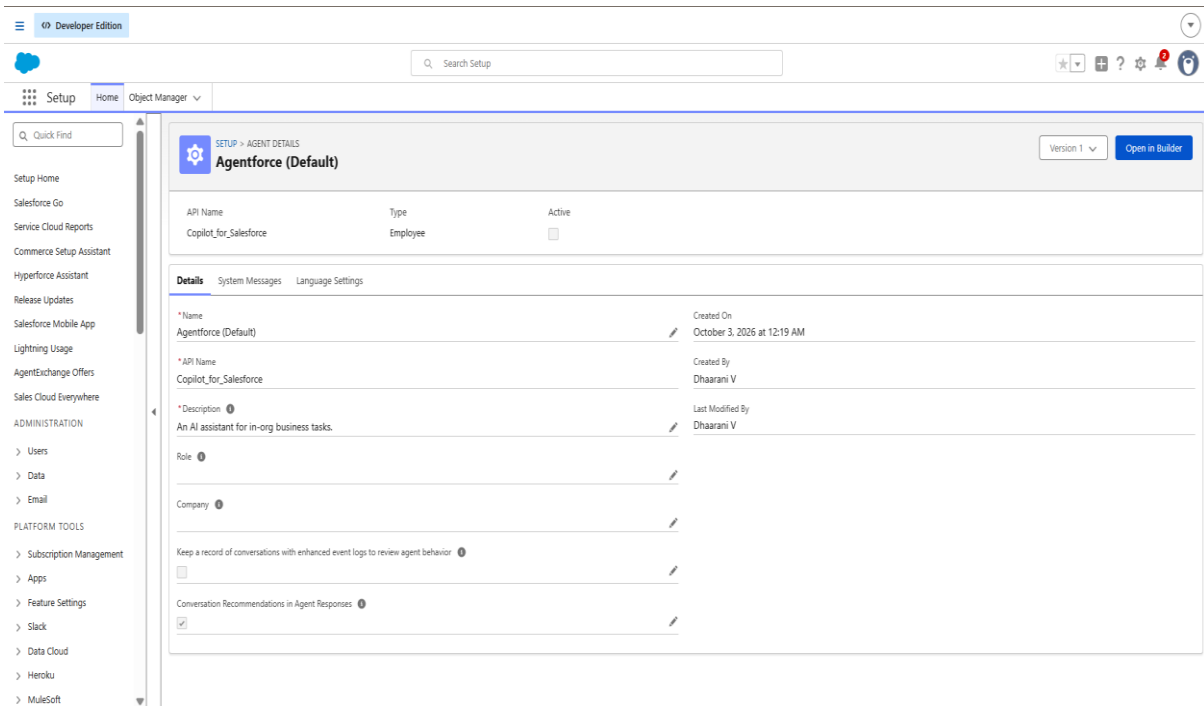
In the Quick Find box on the left, type Users.

1. Click Users under the Users section.
2. Open the Agent User Record

From the list of users, find the Einstein Service Agent user to represent your Agent

(e.g., “SwiftShip Agent”).

Click on the Full Name of that user to open the record.



Milestone -3 Org Setup for AgentForce

Enable salesforce Default agent from Salesforce Go

1. From Quick find box → search “Salesforce Go” → Click on it
2. Search for Default agent → click get started → and turn on
3. And refresh the page

Enable Einstein Setup

Go to Setup → Quick Find → Einstein Setup.

1. Open Einstein Setup → click Get Started if prompted.

Toggle On these key services:

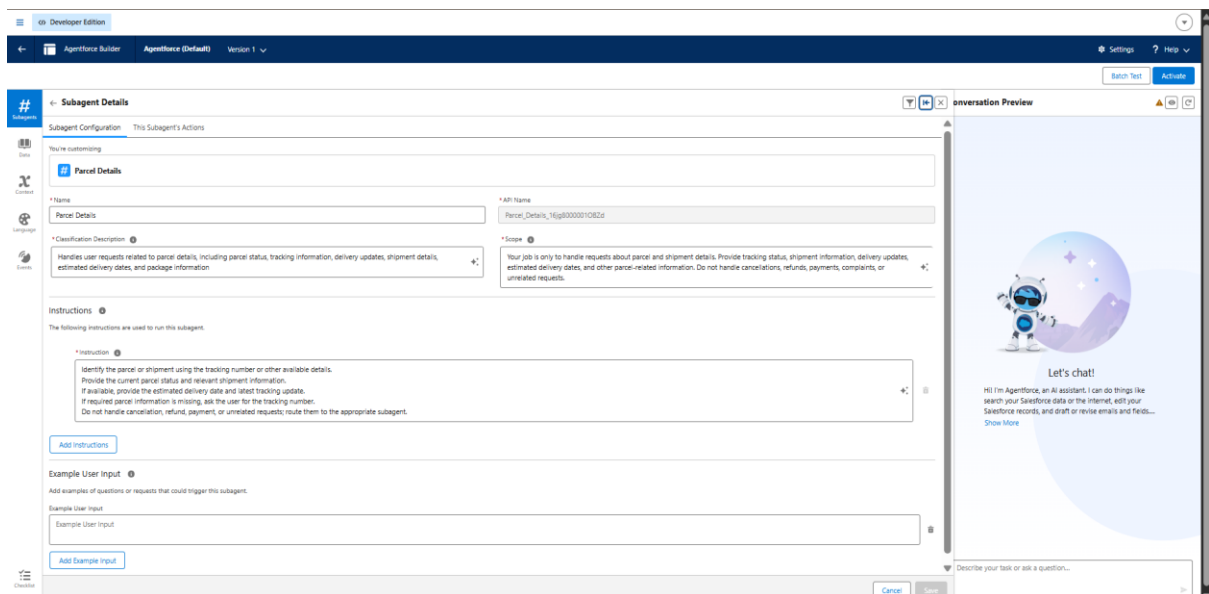
- Turn On Einstein
- Global Languages
- Deploy Prompt Template

3. Refresh your browser page.

4. Go to Setup → Quick Find → AgentForce Agents.

Toggle On these key services:

- Agentforce
- Enable Agentforce Default Agent

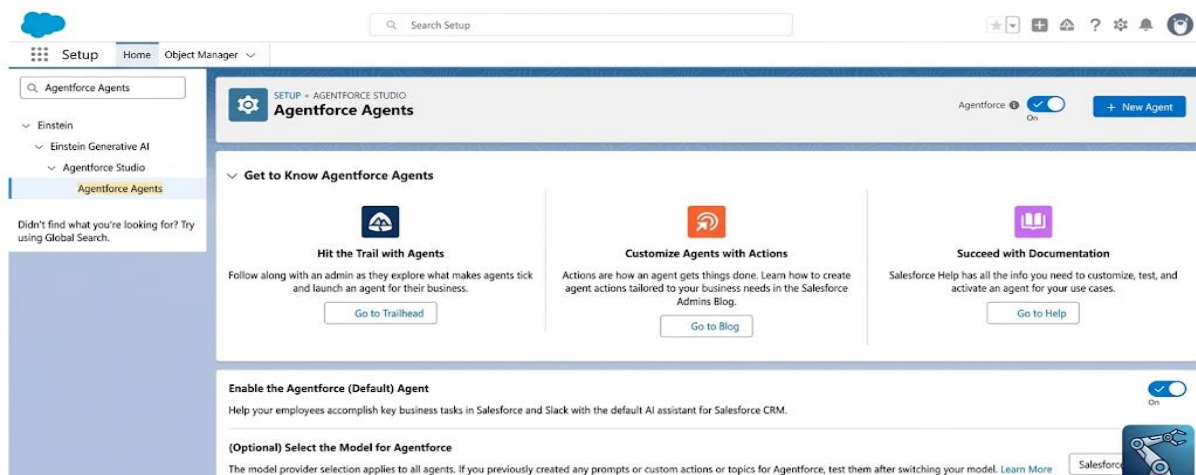


3. Refresh your browser page.

4. Go to Setup → Quick Find → AgentForce Agents.

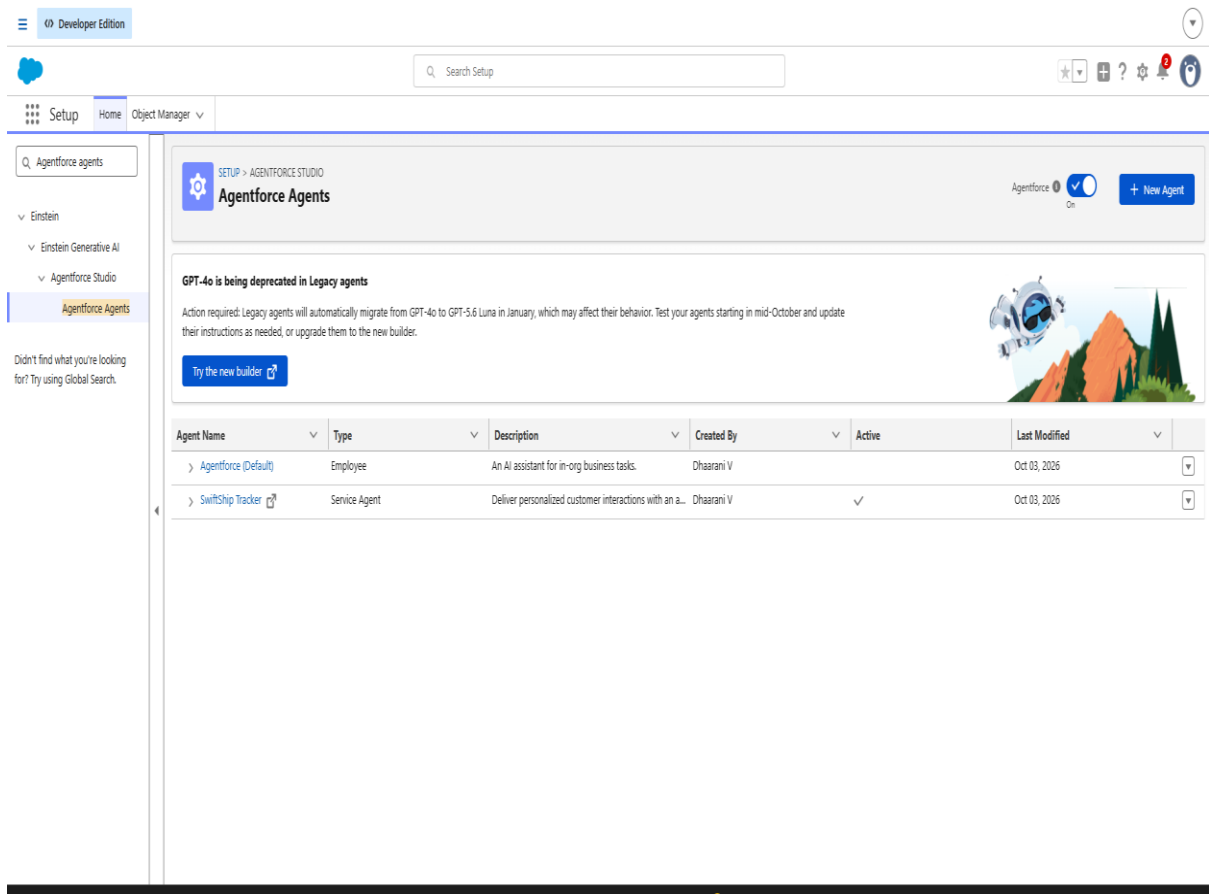
Toggle On these key services:

- Agentforce
- Enable Agentforce Default Agent



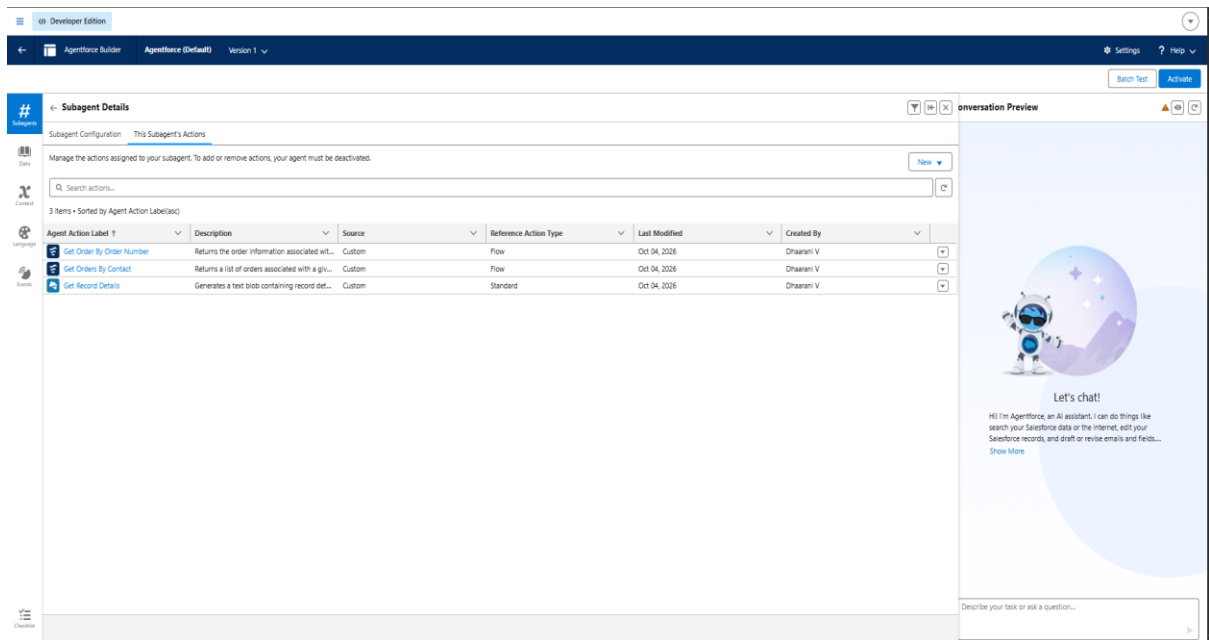
Milestone - 4 Building The Agent

1. In Setup, search for Agentforce Agents in Quick Find.
2. Click Agentforce Agents, then click on the default Agent, Then click on Open in Builder.



The screenshot displays the Salesforce Agentforce Agents setup interface. The top navigation bar includes the 'Developer Edition' label and a search bar. The left sidebar shows the 'Setup' menu with 'Agentforce Agents' selected. The main content area features a header with 'Agentforce Agents' and a 'New Agent' button. Below this, a warning message states 'GPT-4o is being deprecated in Legacy agents' with a 'Try the new builder' button. A table lists the available agents:

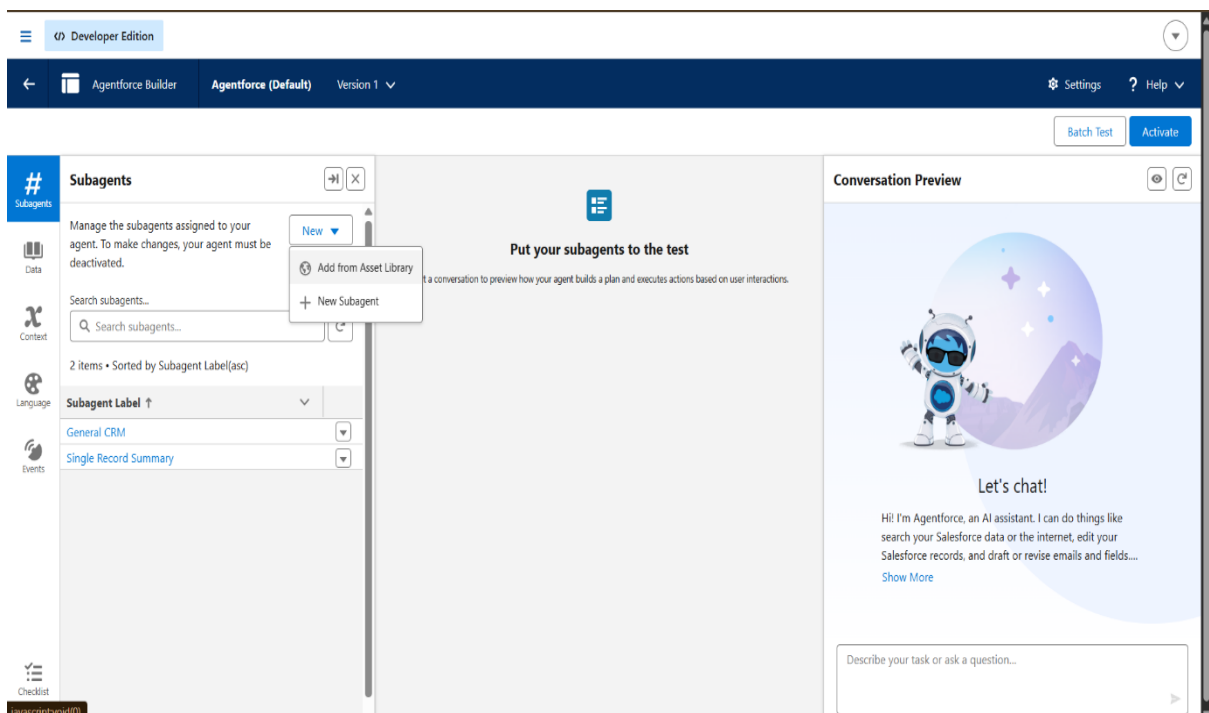
Agent Name	Type	Description	Created By	Active	Last Modified
> Agentforce (Default)	Employee	An AI assistant for in-org business tasks.	Dhaarani V		Oct 03, 2026
> SwiftShip Tracker	Service Agent	Deliver personalized customer interactions with an a...	Dhaarani V	✓	Oct 03, 2026



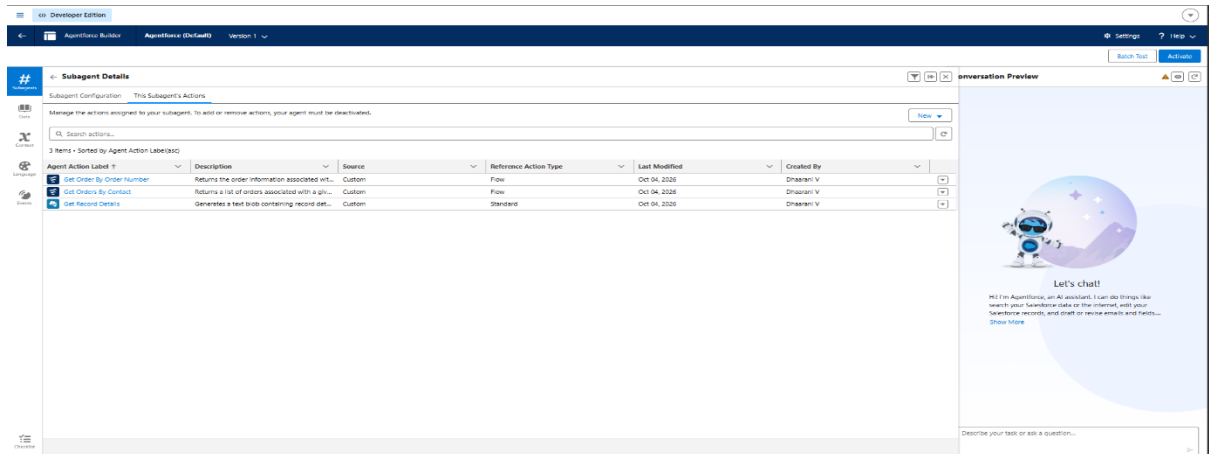
Assign Sub Agents to the Agent

On the subagents Click on New.

1. Click New Subagent



3. Create a new subagent.
4. Fill in the details as shown.



Adding Actions to Agentforce Assets

While still on the Agent Builder in the topics, Click on the Topics Actions

2. Click on New Agent Action.
3. For Action type, Click on Flow then Next.
4. For Reference Flow use the recently created flow, click Next.

The screenshot displays the 'Subagent Details' configuration page in the Salesforce Agentforce Builder. The interface is divided into several sections for configuring a subagent named 'Parcel Details'.

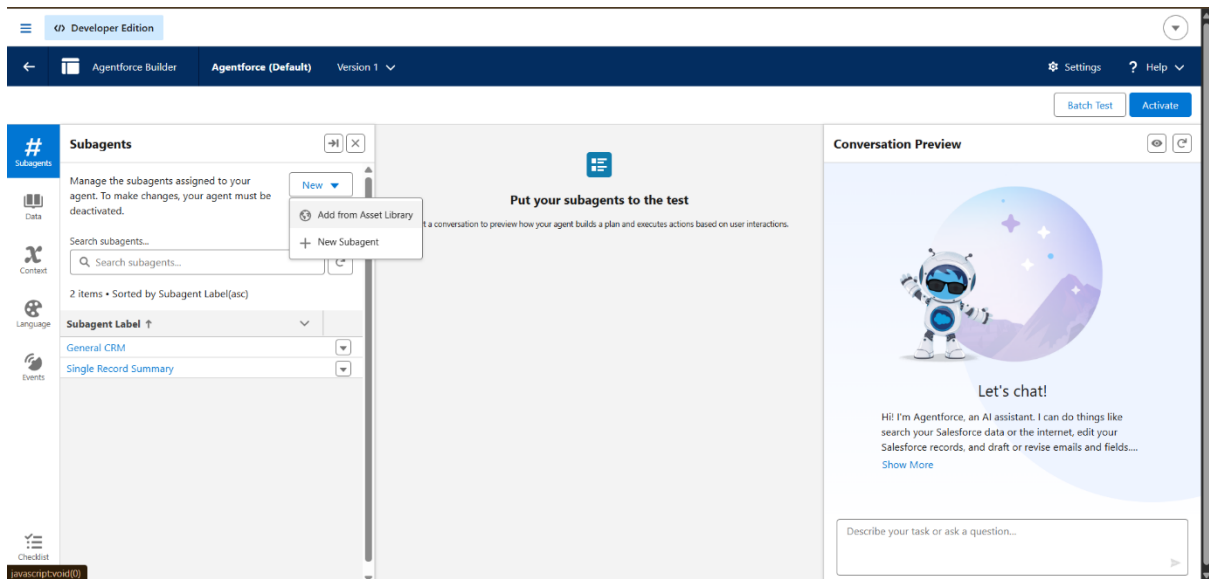
- Subagent Configuration:**
 - Name:** Parcel Details
 - API Name:** Parcel_Details__16ga800001082a
 - Classification Description:** Handles user requests related to parcel details, including parcel status, tracking information, delivery updates, shipment details, estimated delivery dates, and package information.
 - Scope:** Your job is only to handle requests about parcel and shipment details. Provide tracking status, shipment information, delivery updates, estimated delivery dates, and other parcel-related information. Do not handle cancellations, refunds, payments, complaints, or unrelated requests.
- Instructions:**
 - Instruction:** Identify the parcel or shipment using the tracking number or other available details. Provide the current parcel status and relevant shipment information. If available, provide the estimated delivery date and latest tracking update. If required parcel information is missing, ask the user for the tracking number. Do not handle cancellation, refund, payment, or unrelated requests; route them to the appropriate subagent.
- Example User Input:**
 - Example User Input:** Example user input

On the right side, there is an **Conversation Preview** window showing a chat interface with a cartoon robot character. The text in the preview says: "Let's chat! Hi! I'm Agentforce, an AI assistant. I can do things like search your Salesforce data or the Internet, edit your Salesforce records, and draft or revise emails and fields..."

5. Fill the details for Agent Configurations.

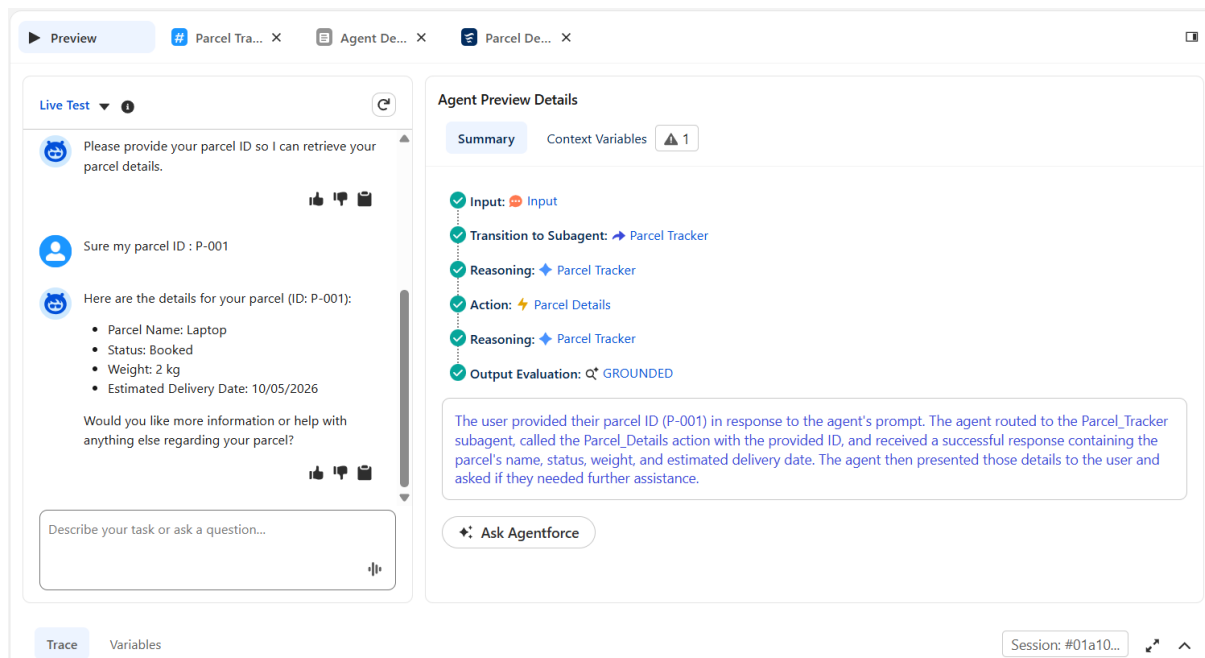
Assign Subagents Action

While on the Agent Builder in Subagent, Click on the Parcel Updates Subagent



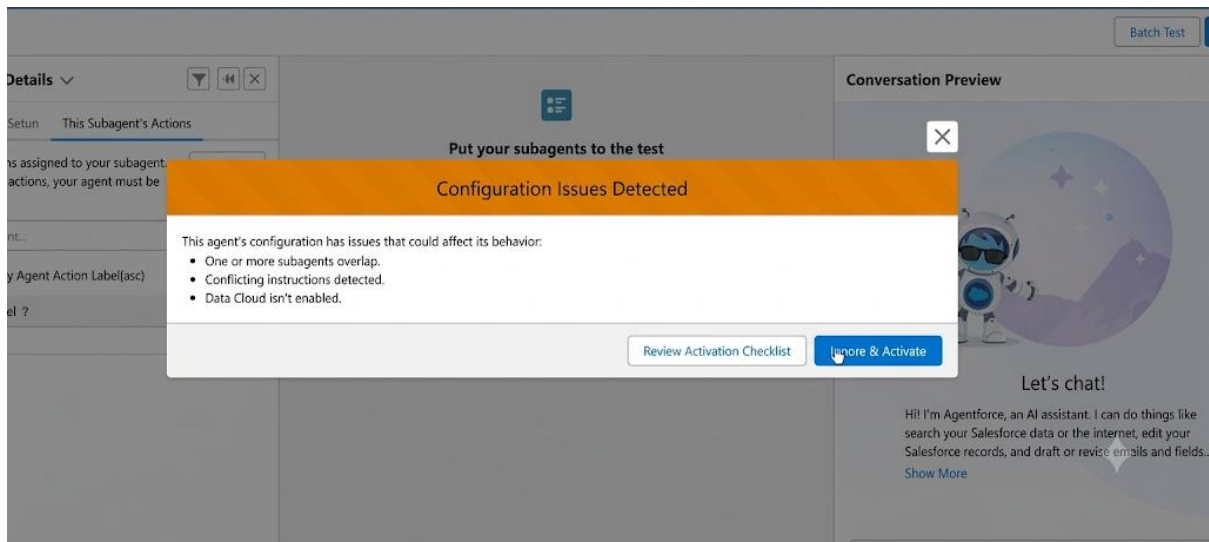
2. Click on the subagents Actions, then select Add from asset library.

3. Select the flow action that we created and click on finish.



4. Click on settings , and rename your agent to SwiftShip Tracker

5. Click Save.



6. Click on Ignore and Activate, then Activate.

Phase 4: Data Migration, Testing & Security

To validate the functionality and reliability of the Parcel Tracking Agent created in Agent Builder. This

phase ensures that the agent correctly recognizes topics, triggers the assigned Flow (SwiftShip

Tracker), retrieves or updates parcel data, and performs all intended actions without access or

permission issues. The goal is to confirm that the AI-driven agent operates seamlessly within

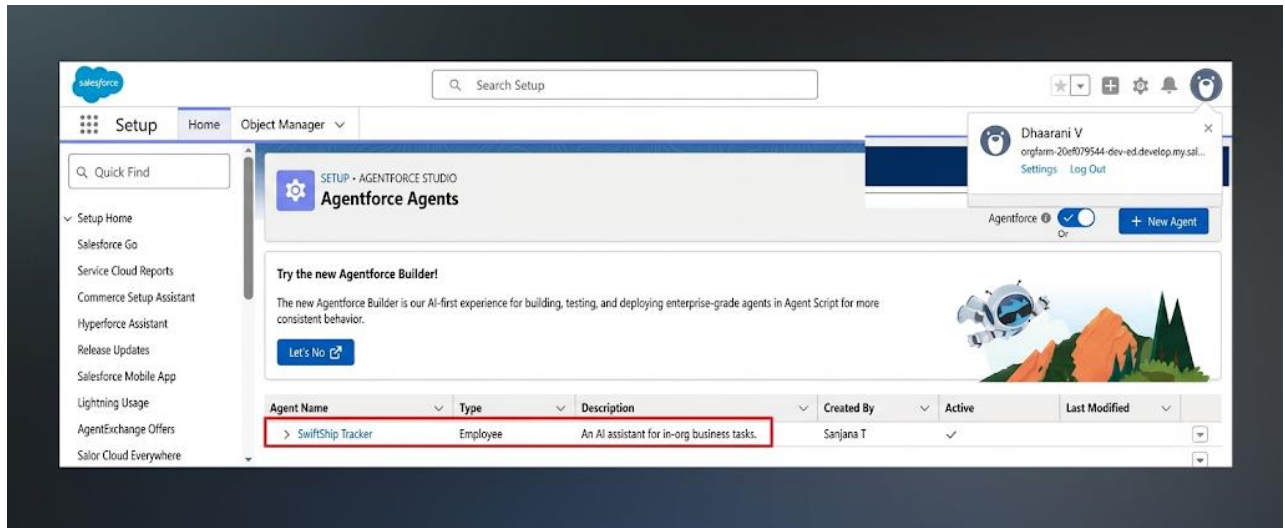
Salesforce and provides accurate, real-time responses to parcel management queries.

Verify Agent

Go to Setup → Agentforce Agents.

1. Open the Agents list and confirm that your agent

Name: Parcel Tracking Agent, appears



Validate Assigned Topic

In Agent Builder → SubAgent, confirm that the SubAgent you created (e.g., Parcel Tracking or

SwiftShip Tracker) is listed.

Open the topic to verify:

- It contains a prompt or description related to parcel management.
- The Action/Flow (SwiftShip Tracker) is linked to this topic.

2. Click Save and ensure the topic is Active.

Preview

Parcel Tra... XAgent De... XParcel De... X

Live Test

Please provide your parcel ID so I can retrieve your parcel details.

Sure my parcel ID : P-001

Here are the details for your parcel (ID: P-001):

- Parcel Name: Laptop
- Status: Booked
- Weight: 2 kg
- Estimated Delivery Date: 10/05/2026

Would you like more information or help with anything else regarding your parcel?

Describe your task or ask a question...

Trace

Variables

Session: #01a10...

Agent Preview Details

Summary

Context Variables 1

Input: Input

Transition to Subagent: Parcel Tracker

Reasoning: Parcel Tracker

Action: Parcel Details

Reasoning: Parcel Tracker

Output Evaluation: GROUNDED

The user provided their parcel ID (P-001) in response to the agent's prompt. The agent routed to the Parcel_Tracker subagent, called the Parcel_Details action with the provided ID, and received a successful response containing the parcel's name, status, weight, and estimated delivery date. The agent then presented those details to the user and asked if they needed further assistance.

Ask Agentforce

Developer Edition

Agentforce BuilderSwiftShip TrackerVersion 1 (Active)

Subagents

Data

Context

Language

Events

Subagent Details

Subagent Configuration

This Subagent's Actions

Manage the actions assigned to your subagent. To add or remove actions, your agent must be deactivated.

Search actions...

1 items • Sorted by Agent Action Label(asc)

Agent Action Label

Parcel Details

Put your subagents to the test

Start a conversation to preview how your agent builds a plan and executes actions based on user interactions.

Check the Flow Association

Go to Setup → Flows.

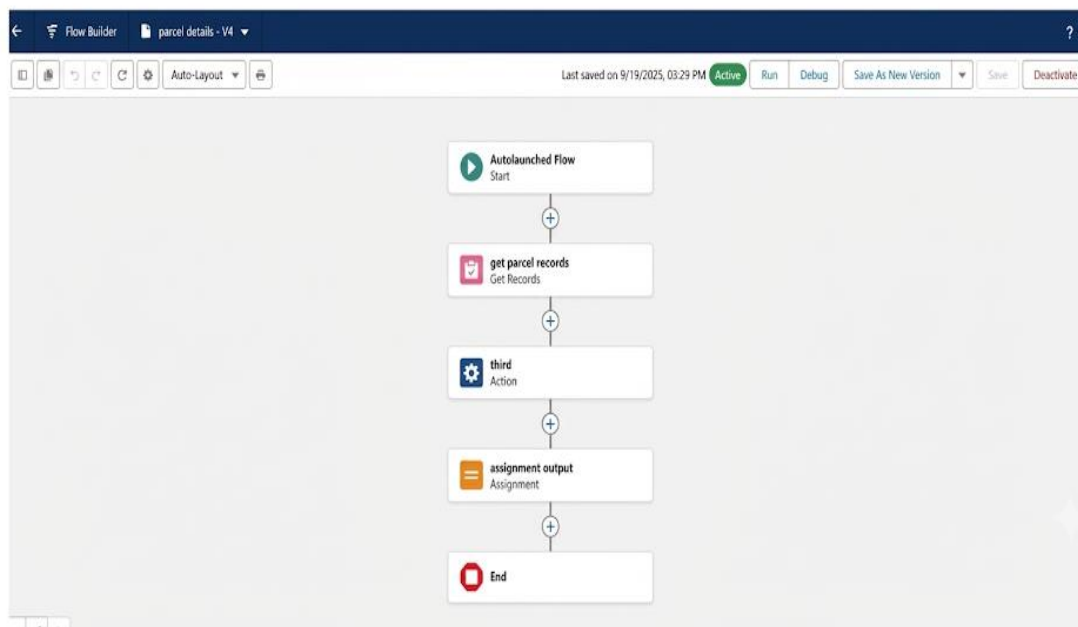
1. Locate the Flow named SwiftShip Tracker.

2. Confirm:

- The Flow is Activated.
- The Flow includes screens or actions that handle:
 - Parcel ID input
 - Parcel details retrieval
 - Status update or confirmation message

3. Open the Flow's Advanced Settings → ensure Run in System Context (Without Sharing) if

data access is restricted.



Activate and Test the Agent

1. Return to Setup → Agent Builder → Parcel Tracking Agent.
2. Click Activate (if not already activated).
3. Once active, test using Preview / Try it Out mode:

Type a query such as:

“Track parcel ID P12345” or

“Update delivery status for parcel P67890 to Delivered.”

4. Observe the agent’s response.

Validate Data in Salesforce

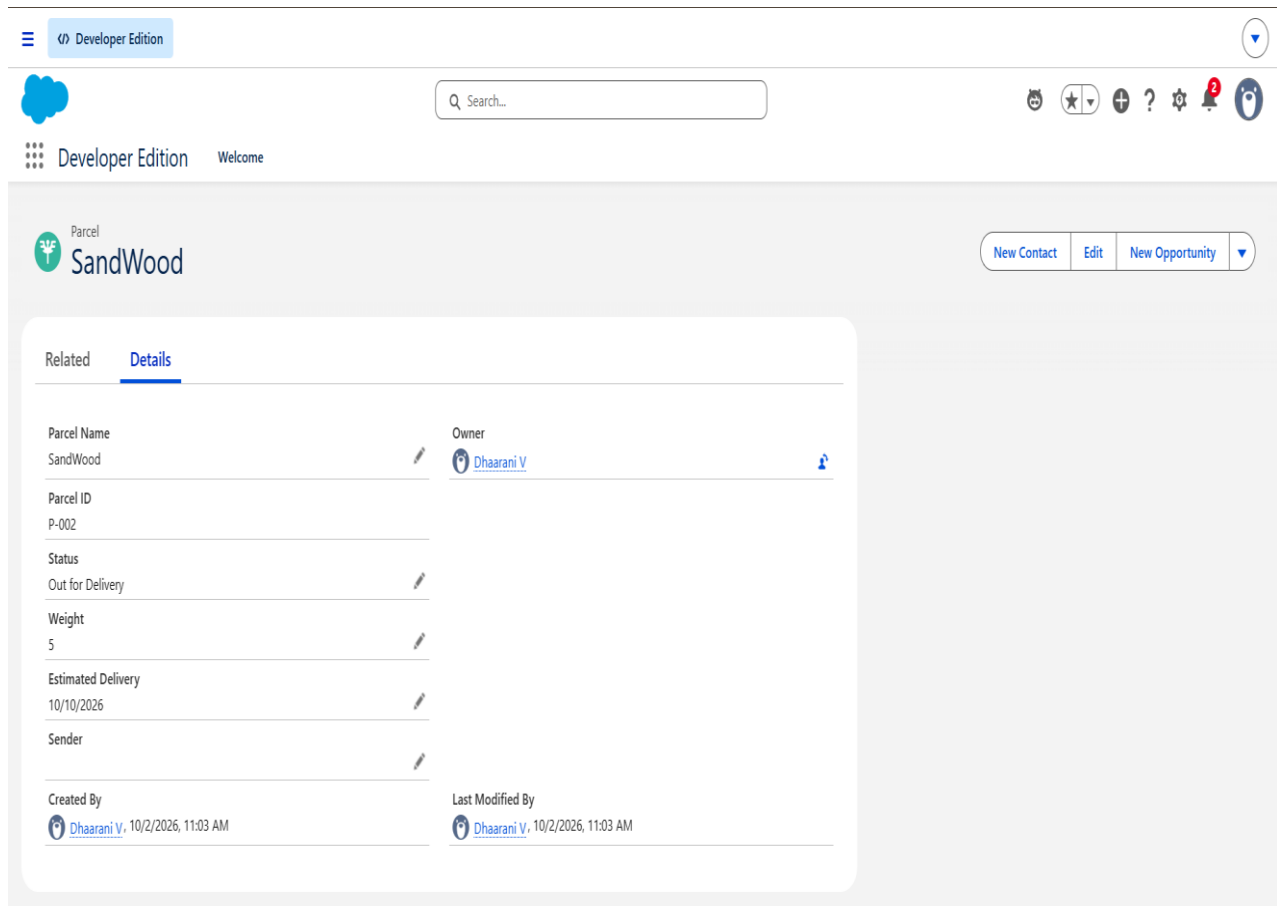
1. After the agent runs the Flow:

- Go to App Launcher → Parcels (Parcel__c).

Search for the Parcel ID you used during testing.

Verify:

- The Status or related field is updated.
- Any Flow-created records (bookings, updates) appear correctly.



Phase 5: Deployment, Documentation & Maintenance

Deployment & Validation

- In the scope of our **Swift Ship Tracker** project implementation:
- The purpose of this phase is to validate that the **Shipment Operations Assistant**, created using Salesforce AgentForce and Prompt Builder, works as intended within the Salesforce Developer Org environment.
- The main goal is to verify that the AI-powered assistant can correctly recognize operational queries, access the appropriate Salesforce information, and provide relevant responses related to **shipments, delivery attempts, routes, and delivery exceptions**.
- Testing is conducted within the Salesforce Developer Org to simulate how the AI-assisted solution would perform in an operational environment, ensuring that **permissions, Flows, data relationships, and AI interactions** function correctly before any deployment stage.
- The validation process includes testing:

- Shipment information retrieval
- Delivery-attempt information retrieval
- Delivery-exception information retrieval
- Exception status and resolution information
- AgentForce topic and action recognition
- Prompt Builder responses
- Salesforce Flow execution
- User access and security permissions
- Reports and dashboard data visibility
- Real-world production testing such as UAT or sandbox deployment is not performed in this project because:
 - Salesforce Developer Edition environments are standalone development environments and are not used as production-linked enterprise testing environments.
 - Enterprise-level Salesforce deployment involves sandbox testing, staging environments, deployment pipelines, and CI/CD automation, which are outside the scope of this academic/prototype implementation.

Maintenance, Monitoring & Troubleshooting

In the **Swift Ship Tracker** project, ongoing maintenance ensures that **Shipment, Route, Delivery Attempt, Delivery Exception, and Proof of Delivery** data remain accurate, consistent, and up to date.

Regular monitoring helps identify issues such as incorrect shipment information, failed Flow executions, incorrect exception assignments, unsuccessful automation, permission-related problems, or delays in AgentForce responses.

Troubleshooting activities include:

- Resolving Salesforce configuration errors.
- Updating Flow logic when operational requirements change.
- Correcting data relationship or validation issues.
- Refining AgentForce topics and actions.
- Improving Prompt Builder instructions and responses.

- Reviewing failed automation and notification processes.
- Updating security permissions when user responsibilities change.
- Monitoring reports and dashboards for incorrect or incomplete operational data.
- Maintaining Apex logic when business rules or shipment-processing requirements change.

Although this project is implemented within a **Salesforce Developer Org for academic and learning purposes**, these maintenance and monitoring practices would be important in a real-world logistics deployment.

They help ensure that the shipment management system remains **reliable, scalable, secure, and aligned with changing logistics operations**.

Continuous improvement of AgentForce topics, Prompt Builder instructions, Salesforce Flows, security controls, reports, and operational data can further improve the effectiveness of the Swift Ship Tracker solution.

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to validate **shipment creation, route assignment, delivery-attempt recording, delivery exception management, exception escalation, proof-of-delivery processing, automated notifications, AgentForce responses, and dashboard analytics**.

All major features were tested to verify that the SwiftShip Tracker functions correctly according to the defined requirements.

Testing Areas

- **Shipment Creation Testing**

Verified that valid shipment records can be created with the required information.

- **Route Assignment Testing**

Verified that shipments can be associated with appropriate routes and assigned delivery agents.

- **Shipment Status Testing**

Verified that shipment milestones can be updated throughout the delivery lifecycle.

- **Delivery Attempt Testing**

Verified that delivery agents can record delivery attempts and maintain attempt history.

- **Delivery Exception Testing**

Verified that delivery problems can be recorded with appropriate exception type, severity, and resolution status.

- **Exception Automation Testing**

Verified that high-priority or critical exceptions trigger the configured assignment, follow-up task, and notification processes.

- **Proof of Delivery Testing**

Verified that delivery completion information can be recorded against the appropriate shipment.

- **AgentForce Testing**

Tested conversational queries related to shipment status, delivery attempts, and unresolved exceptions.

- **Prompt Builder Testing**

Verified that configured prompts can retrieve relevant shipment and exception information and generate appropriate responses.

- **Reports & Dashboard Testing**

Verified that shipment, route, exception, and delivery information is correctly represented in reports and dashboards.

- **Security Testing**

Verified that users can access only the records and fields permitted according to their roles, profiles, permission sets, and security configuration.

7.1 Functional Test Cases

Test ID	Test Scenario	Test Action	Expected Result	Actual Result	Status
TC-01	Shipment Creation	Create a shipment with valid details	Shipment record is created successfully	Record created successfully	Pass
TC-02	Route Assignment	Assign a route and delivery agent to a shipment	Route and agent are associated with shipment	Assignment displayed correctly	Pass
TC-03	Shipment Status	Update shipment milestone	Shipment status changes to selected milestone	Status updated correctly	Pass
TC-04	Delivery Attempt	Create a delivery attempt	Attempt is stored and linked to shipment	Record created successfully	Pass
TC-05	Delivery Exception	Create a high-priority exception	Exception is created with correct severity and status	Exception displayed correctly	Pass
TC-06	Exception Automation	Create a high/critical exception	Configured Flow performs assignment/follow-up/notification	Automation executed successfully	Pass
TC-07	Proof of Delivery	Record successful delivery confirmation	Proof-of-delivery record is associated with shipment	Record created successfully	Pass
TC-08	Agentforce	Ask an operational shipment query	Agentforce returns relevant shipment information	Response generated	Pass
TC-09	Prompt Builder	Test shipment summary prompt	Prompt generates response using	Response generated	Pass

Test ID	Test Scenario	Test Action	Expected Result	Actual Result	Status
			configured Salesforce data		
TC-10	Reports & Dashboard	Open reports and dashboard	Current shipment and exception data is displayed	Data displayed correctly	Pass
TC-11	Security	Access records using appropriate user permissions	User can access only permitted information	Access behavior verified	Pass

7.2 Performance & Reliability Checks

Test ID	Check	Expected Result	Status
PC-01	Lightning App loading	Application opens without an unexpected error	Pass
PC-02	Record Save	Shipment/exception records save successfully	Pass
PC-03	Flow Execution	Configured automation completes without an error	Pass
PC-04	Related Records	Related shipment records are displayed correctly	Pass
PC-05	Agentforce Response	Agentforce returns a relevant response for the tested query	Pass
PC-06	Dashboard Refresh	Dashboard reflects available Salesforce data correctly	Pass
PC-07	Security	Unauthorized access is restricted according to configuration	Pass

Build the required UI

Step 1 — Create the custom objects

Go to:

Setup → Object Manager → Create → Custom Object

Create:

1. Shipment

Label:

Shipment

API:

Shipment__c

Recommended fields:

- Shipment Number
- Tracking Code
- Shipment Status
- Priority
- Origin
- Destination
- Dispatch Date
- Expected Delivery Date
- Current Milestone
- Exception Flag

The screenshot shows the 'New Custom Object' setup page in Salesforce Developer Edition. The page is titled 'New Custom Object' and includes a 'Custom Object Definition Edit' section. The 'Custom Object Information' section contains the following fields:

- Label:** Shipment (Example: Account)
- Plural Label:** Shipments (Example: Accounts)
- Starts with vowel sound:** ☐
- Object Name:** Shipment (Example: Account)
- Description:** (Empty text area)

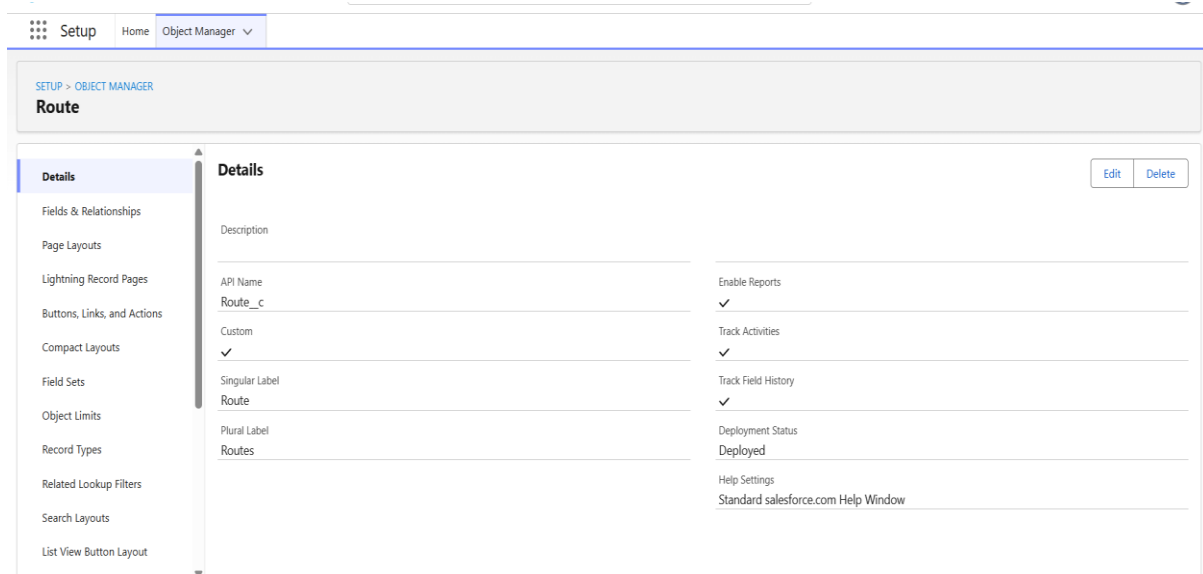
At the bottom of the page, there are links for 'Context Sensitive Help', 'Settings', and 'Open the standard Salesforce.com Help & Training window'. A right-hand sidebar shows the user's profile (kavya V) and navigation options like 'Settings', 'Log Out', 'DISPLAY DENSITY', 'Compact', and 'Switch to Salesforce Classic'.

2. Route

Route__c

Fields:

- Route Name
- Route Code
- Origin
- Destination
- Distance
- Route Status
- Assigned Delivery Agent
- Planned Start
- Planned End



3. Delivery Attempt

Delivery_Attempt__c

Fields:

- Shipment
- Delivery Agent

- Attempt Date
- Attempt Number
- Attempt Status
- Remarks
- Recipient Availability
- Next Attempt Date

SETUP > OBJECT MANAGER

Delivery Attempt

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

Details

Description

API Name

Delivery_Attempt__c

Custom

✓

Singular Label

Delivery Attempt

Plural Label

Delivery Attempts

Enable Reports

✓

Track Activities

✓

Track Field History

✓

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

Edit Delete

4. Delivery Exception

Delivery_Exception__c

Fields:

- Exception Number
- Shipment
- Exception Type
- Severity
- Description

- Reported Date
- Assigned To
- Resolution Status
- Resolution Notes
- Resolution Date
- Escalation Required

The screenshot shows the Salesforce Setup interface for the 'Delivery Exception' object. The breadcrumb trail at the top reads 'SETUP > OBJECT MANAGER'. The page title is 'Delivery Exception'. On the left, a sidebar lists various configuration options: Details (selected), Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, and List View Button Layout. The main content area is titled 'Details' and contains a table of configuration settings. The table has two columns: the first column lists the setting name, and the second column shows its current value or status. The settings are: Description (empty), API Name (Delivery_Exception__c), Custom (checked), Singular Label (Delivery Exception), Plural Label (Delivery Exceptions), Enable Reports (checked), Track Activities (checked), Track Field History (checked), Deployment Status (Deployed), and Help Settings (Standard salesforce.com Help Window). In the top right corner of the main content area, there are 'Edit' and 'Delete' buttons.

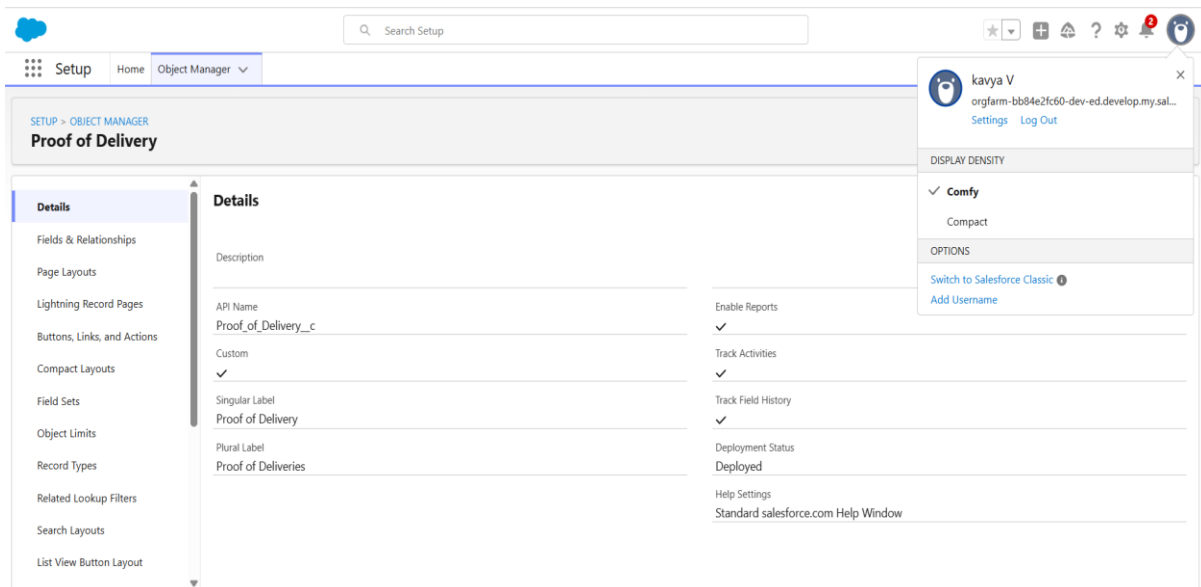
Setting	Value
Description	
API Name	Delivery_Exception__c
Custom	✓
Singular Label	Delivery Exception
Plural Label	Delivery Exceptions
Enable Reports	✓
Track Activities	✓
Track Field History	✓
Deployment Status	Deployed
Help Settings	Standard salesforce.com Help Window

5. Proof of Delivery

Proof_of_Delivery__c

Fields:

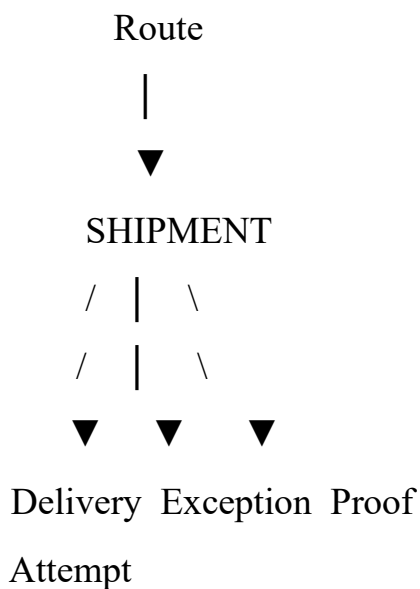
- Shipment
- Delivery Date
- Receiver Name
- Confirmation Type
- Confirmation Status
- Delivery Remarks
- Recorded By



7. Create the relationships

This is extremely important.

Your main relationship should look like:



So:

Shipment → Route

Use a relationship field.

Delivery Attempt → Shipment

Use Lookup or Master-Detail depending on your design.

Delivery Exception → Shipment

Lookup/Master-Detail.

Proof of Delivery → Shipment

Lookup/Master-Detail.

This lets you demonstrate related records in the Salesforce UI.

8. Build the SwiftShip Tracker Lightning App

Go to:

Setup → App Manager → New Lightning App

Name:

SwiftShip Tracker

Add navigation items:

Home

Shipments

Routes

Delivery Attempts

Delivery Exceptions

Proof of Delivery

The screenshot shows the Salesforce Lightning Experience App Manager interface. The top navigation bar includes the Setup menu, Home link, and Object Manager dropdown. A search bar is present in the top right. The left sidebar shows the App Manager menu item selected under the Apps section. The main content area displays a table of 32 items, sorted by App Name. The table columns are App Name, Developer Name, Description, Last Modified Date, App Type, and Visibility. The SwiftShip Tracker app is listed at the bottom of the table.

	App Name	Developer Name	Description	Last Modified Date	App Type	Visibl...
21	Sales	Sales	The world's most popular sales force automation (SFA) solution	9/24/2026, 10:42 AM	Classic	
22	Sales	LightningSales	Manage your sales process with accounts, leads, opportunities, and more	9/24/2026, 10:42 AM	Lightning	✓
23	Sales Cloud Mobile	SalesCloudMobile	New seller focused mobile first experience	9/24/2026, 10:42 AM	Lightning	✓
24	Sales Console	LightningSalesCons...	(Lightning Experience) Lets sales reps work with multiple records on one scre...	9/24/2026, 10:42 AM	Lightning	✓
25	Salesforce Chatter	Chatter	The Salesforce Chatter social network, including profiles and feeds	9/24/2026, 10:42 AM	Classic	✓
26	Salesforce Schedule...	LightningScheduler	Set up personalized appointment scheduling.	9/24/2026, 10:45 AM	Lightning	✓
27	Service	Service	Manage customer service with accounts, contacts, cases, and more	9/24/2026, 10:42 AM	Classic	✓
28	Service Console	LightningService	(Lightning Experience) Lets support agents work with multiple records across...	9/24/2026, 10:42 AM	Lightning	✓
29	Shield	Shield	Monitor your Salesforce apps and data. Discover, track, and encrypt sensitive...	9/24/2026, 10:42 AM	Lightning	✓
30	Site.com	Sites	Build pixel-perfect, data-rich websites using the drag-and-drop Site.com app...	9/24/2026, 10:42 AM	Classic	
31	Subscription Mana...	RevenueCloudCons...	Get started automating your revenue processes	9/24/2026, 10:42 AM	Lightning	✓
32	SwiftShip Tracker	SwiftShip_Tracker		10/4/2026, 7:11 AM	Lightning	✓

Test Case — Shipment Creation

SwiftShip Tracker

Home Shipments Tracking History Reports Dashboards

Shipment SHP-1001

Origin: Chennai, Destination: Thiruvapur, Priority: High, Expected Delivery: 08-Oct-2026

Created

Details

System Information

Shipment Number: SHP-1001, Tracking Code: SWT1001, Status: Created

Shipment Information

Origin: Chennai, Destination: Thiruvapur, Expected Delivery: 08-Oct-2026, Created By: Admin User, 08-Oct-2026

Owner: Admin User

Related Lists

- Activity (0)
- Tracking History (1)
- Related Cases (0)
- Files (0)
- Notes (0)

Test Case — Route Assignment

SwiftShip Tracker

Home Shipments Tracking History Reports Dashboards

Shipment SHP-1001

Origin: Chennai, Destination: Thiruvapur, Priority: High, Expected Delivery: 08-Oct-2026

Save

Confirming Assignment

Details **Route Assignment**

System Information

Shipment Number: SHP-1001, Assign Route: CHN-TRR-01, Assign Delivery Agent: Agent-123 (J. Doe), Status: Created

Shipment Information

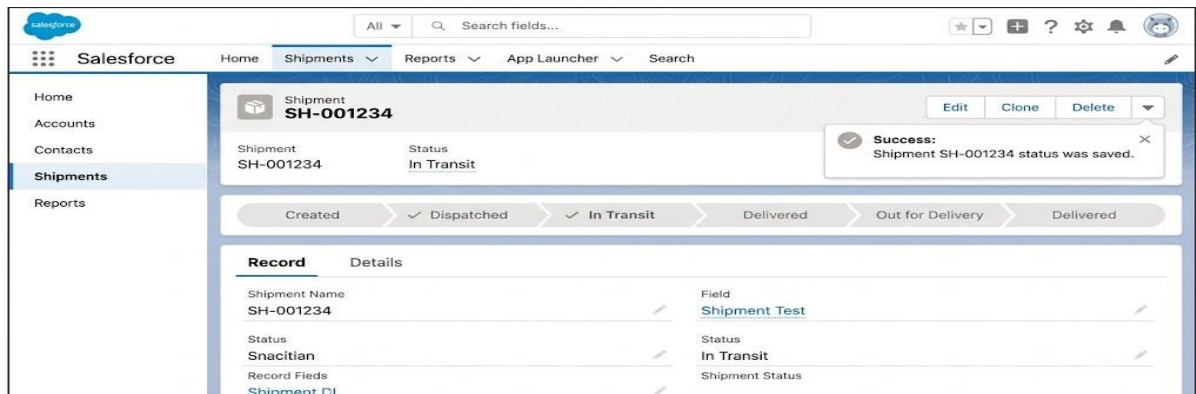
Origin: Chennai, Destination: Thiruvapur, Expected Delivery: 08-Oct-2026, Created By: Admin User, 08-Oct-2026

Owner: Admin User

Related Lists

- Activity (0)
- Tracking History (1)
- Related Cases (0)
- Files (0)
- Notes (0)

Test Case — Shipment Status



Test Case — Exception Automation

Now this is where Salesforce Flow becomes important.

Suppose you create a Flow:

Delivery Exception Created



Check Severity



Is Severity = High/Critical?



YES



Assign Operations User



Create Follow-up Task



Send Email Alert

Security Testing

You also mentioned Security Testing in your Section 7.

Create/test different users or permission sets if your environment allows it.

For example:

Delivery Agent

Should be able to:

- ✓ View assigned shipment
- ✓ Update delivery attempt
- ✓ Create relevant exception

Should not have unrestricted administrative access.

Operations Manager

Should be able to:

- ✓ View shipments
- ✓ View exceptions
- ✓ View reports
- ✓ View dashboard
- ✓ Manage operational records

Admin

Should have configuration access.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralizes shipment, route, delivery-attempt, exception, and proof-of-delivery information in Salesforce.
- Provides better visibility into shipment progress and operational activities.
- Helps operations teams identify delayed and problematic shipments.
- Provides structured delivery-exception management.
- Automates exception assignment, follow-up activities, and operational notifications using Salesforce Flow.

- AgentForce provides conversational assistance for shipment and exception-related queries.
- Prompt Builder supports AI-powered shipment summaries and operational responses.
- Maintains a structured history of delivery attempts.
- Provides centralized proof-of-delivery information.
- Reports and dashboards improve monitoring of shipment and delivery performance.
- Role-based security helps control access to operational information.
- Provides a scalable foundation for integration with external logistics systems.

Disadvantages

- Requires accurate and up-to-date shipment and operational data.
- Depends on correct Salesforce configuration and automation.
- AgentForce responses depend on properly configured topics, actions, prompts, and available Salesforce data.
- Complex Flow and Apex logic may require regular maintenance.
- External logistics integrations may increase implementation complexity.
- Users require appropriate Salesforce permissions to access operational information.
- Initial Salesforce configuration and development can require significant effort.
- Large-scale shipment operations may require additional optimization and architecture.
- Expanding the solution to external GPS, carrier, or logistics systems may require further customization.

9. CONCLUSION

The **SwiftShip Tracker** demonstrates how Salesforce can be used to develop a centralized **shipment operations and delivery exception management solution**.

The project integrates custom objects, relationships, Salesforce Flows, Apex, AgentForce AI, Prompt Builder, security controls, reports, and dashboards to support shipment creation, route coordination, delivery-attempt tracking, exception management, proof of delivery, and operational monitoring.

The integration of **AgentForce AI and Prompt Builder** provides authorized operations users with a conversational method of obtaining shipment and exception information. Salesforce Flow further reduces manual operational effort by automating status updates, exception assignments, follow-up activities, and notifications.

The project therefore provides a practical and scalable foundation for improving shipment visibility, managing delivery exceptions, and supporting logistics operations through the Salesforce platform.

10. FUTURE SCOPE

- Extend AgentForce capabilities to support additional shipment and operational queries.
- Introduce real-time GPS and vehicle-location integration.
- Implement predictive analytics for identifying potential shipment delays.
- Add advanced route-performance and delivery-efficiency analytics.
- Integrate external courier and logistics systems through APIs.
- Introduce a mobile application for delivery agents.
- Enhance digital proof-of-delivery capabilities using signatures or digital evidence.
- Improve AI prompts and conversational workflows for exception management.
- Introduce automated escalation for long-pending critical exceptions.
- Implement enterprise-level sandbox and CI/CD deployment practices.
- Expand automation for additional shipment and logistics operations.

- Develop a customer-facing portal in a future version for shipment-status visibility.

The future development of the system can focus on **AI enhancement, predictive shipment analysis, external logistics integration, mobile delivery operations, real-time tracking, and enterprise deployment.**

11. APPENDIX

A. Custom Objects

- Shipment__c
- Route__c
- Delivery_Attempt__c
- Delivery_Exception__c
- Proof_of_Delivery__c

B. Standard Objects

- Account
- Contact
- Case
- Task
- User

C. Automation

- Record-Triggered Flows
- Auto-Launched Flows
- Scheduled Flows
- Email Alerts
- Operational Tasks

D. Apex

- Apex Classes
- Batch Apex, where required

E. AI

- AgentForce AI
- Prompt Builder

- Shipment Operations Assistant
- Exception Summary

F. UI

- Lightning App Page
- Lightning Record Pages
- Dynamic Forms
- Related Lists

G. Security

- Profiles
- Roles
- Permission Sets
- Sharing Rules
- Field-Level Security

H. Analytics

- Shipment Reports
- Delivery Exception Reports
- Route Performance Reports
- Delivery Attempt Reports
- Operational Dashboards

I. Testing

- Shipment Creation Testing
- Route Assignment Testing
- Delivery Attempt Testing
- Exception Management Testing
- Flow Automation Testing
- AgentForce Testing
- Prompt Builder Testing
- Report & Dashboard Validation
- Security and Permission Validation

THANK YOU